

Review

Visual and Visual–Inertial SLAM for UGV Navigation in Unstructured Natural Environments: A Survey of Challenges and Deep Learning Advances

Tiago Pereira ^{1,2,3,4,*} , Carlos Viegas ³ , Salviano Soares ^{1,5,6}  and Nuno Ferreira ^{2,7} 

- ¹ Engineering Department, School of Sciences and Technology, University of Trás-os-Montes and Alto Douro (UTAD), Quinta de Prados, 5000-801 Vila Real, Portugal; salblues@utad.pt
 - ² GECAD—Knowledge Research Group on Intelligent Engineering and Computing for Advanced Innovation and Development, Engineering Institute of Porto (ISEP), Polytechnic Institute of Porto (IPP), 4200-465 Porto, Portugal; nunomig@isec.pt
 - ³ Univ Coimbra, ADAI, Department of Mechanical Engineering, Rua Luís Reis Santos, Pólo II, 3030-788 Coimbra, Portugal; carlos.viegas@uc.pt
 - ⁴ Polytechnic University of Coimbra, Rua da Misericórdia, Lagar dos Cortiços, S. Martinho do Bispo, 3045-093 Coimbra, Portugal
 - ⁵ Institute of Electronics and Informatics Engineering of Aveiro (IEETA), University of Aveiro, 3810-193 Aveiro, Portugal
 - ⁶ Intelligent Systems Associate Laboratory (LASI), University of Aveiro, 3810-193 Aveiro, Portugal
 - ⁷ Institute for Systems and Computer Engineering-Technology and Science (INESC TEC), Rua Dr. Roberto Frias, 4200-465 Porto, Portugal
- * Correspondence: tiago.pereira@ipc.pt

Abstract

Localization and mapping remain critical challenges for Unmanned Ground Vehicles (UGVs) operating in unstructured natural environments, such as forests and agricultural fields. While Visual SLAM (VSLAM) and Visual–Inertial SLAM (VI-SLAM) have matured significantly in structured and urban scenarios, their extension to outdoor natural domains introduces severe challenges, including dynamic vegetation, illumination variations, a lack of distinctive features, and degraded GNSS availability. Recent advances in Deep Learning have brought promising developments to VSLAM- and VI-SLAM-based pipelines, ranging from learned feature extraction and matching to self-supervised monocular depth prediction and differentiable end-to-end SLAM frameworks. Furthermore, emerging methods for adaptive sensor fusion, leveraging attention mechanisms and reinforcement learning, open new opportunities to improve robustness by dynamically weighting the contributions of camera and IMU measurements. This review provides a comprehensive overview of Visual and Visual–Inertial SLAM for UGVs in unstructured environments, highlighting the challenges posed by natural contexts and the limitations of current pipelines. Classic VI-SLAM frameworks and recent Deep-Learning-based approaches were systematically reviewed. Special attention is given to field robotics applications in agriculture and forestry, where low-cost sensors and robustness against environmental variability are essential. Finally, open research directions are discussed, including self-supervised representation learning, adaptive sensor confidence models, and scalable low-cost alternatives. By identifying key gaps and opportunities, this work aims to guide future research toward resilient, adaptive, and economically viable VSLAM and VI-SLAM pipelines, tailored for UGV navigation in unstructured natural environments.

Keywords: SLAM; visual SLAM; visual–inertial SLAM; Deep Learning; computer vision; unstructured natural environments



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1. Introduction

The field of robotics has experienced rapid growth in recent years, driving the development of new technologies and applications. Consequently, robotic systems have attracted significant commercial interest due to their ability to assist humans in tasks that are not only repetitive but also hazardous [1–3]. One area in which robotic solutions have seen significant research progress is the integration of Unmanned Ground Vehicles (UGVs) in unstructured natural environments [4], which are outdoor, heterogeneous, and dynamic settings characterized by continuous variations in terrain, vegetation, and lighting, often under adverse and unpredictable conditions [5,6]. These types of environments are closely associated with the agricultural and forestry sectors, both of which show a growing need for process optimization. In agriculture, there is an increasing demand for automated machinery capable of performing tasks such as harvesting, planting, and crop monitoring and inspection, among others. In the forestry sector, the integration of UGVs supports tasks such as preservation, clearing, monitoring of forest areas, and even wildfire mitigation and response [4].

Within this context, the development of solutions enabling UGVs to navigate autonomously in unknown, unstructured natural environments becomes increasingly essential. Accordingly, these advancements impose a fundamental requirement: robots must be capable of localizing themselves and building maps in unfamiliar environments. This requirement is embodied in Simultaneous Localization and Mapping (SLAM) [7], a concept that has attracted increasing research interest in robotics and Artificial Intelligence (AI). SLAM is the process that enables a mobile robot to incrementally build a consistent map of an unknown surrounding environment while simultaneously estimating its own location within that map [8]. The robot model typically includes its position and, in some cases, velocity, sensor parameters, and other relevant characteristics. The map corresponds to a multidimensional representation of the elements detected by the robot's onboard sensors, which serve as reference points for the localization process.

According to the predominant sensors employed, SLAM can be categorized as Visual SLAM (VSLAM) or LiDAR-SLAM (LSLAM) [9]. LSLAM relies on LiDAR (Light Detection and Ranging) sensors to generate a detailed three-dimensional map of an environment while simultaneously estimating the sensor's pose within that map [10]. VSLAM, on the other hand, uses images to interpret the environment, providing a perception similar to human vision and offering richer information when compared with LSLAM [11]. In recent years, research on VSLAM has expanded significantly, attracting substantial attention due to its accessibility, ease of installation, and ability to deliver extensive information about the surrounding environment. Moreover, VSLAM can be fused with other sensors, contributing to the advancement of a wide range of algorithms. For UGV navigation, VSLAM has become a key solution because of its intuitive environment perception based on geometric image features. The maps generated are particularly valuable in agro-forestry sectors, where workers benefit from detailed spatial information about their fields.

However, cameras tend to fail when used alone in complex scenarios such as agricultural or forest environments [7]. For this reason, VSLAM is often fused with other sensors such as LiDAR, Inertial Measurement Units (IMUs), and Global Navigation Satellite Systems (GNSSs). Among the several possible combinations, the most widely adopted and low-cost paradigm is Visual-Inertial SLAM (VI-SLAM) [12].

Despite the significant advances in VSLAM, these systems still face limitations when deployed in unstructured natural environments. Variations in illumination, dynamic vegetation, irregular terrain, and degraded GNSS signals in forested areas directly affect the robustness and accuracy of existing algorithms. The integration of Deep Learning techniques has been transforming the VSLAM landscape by introducing improvements in visual feature extraction (e.g., SuperPoint [13], LightGlue [14]), depth estimation (e.g., MonoDepth2 [15], end-to-end pipelines (e.g., DROID-SLAM [16] and SuperVINS [17]) and semantic mapping, which enables robots to acquire enhanced scene perception and understanding [18].

Despite the various solutions already proposed to address localization and mapping challenges in agricultural and forestry environments, it is evident that additional research is still required. This complexity has driven the emergence of new methodologies and the evolution of concepts aimed at improving robot autonomy in outdoor environments using VSLAM as a foundational technology.

While a number of surveys and reviews have explored vision-based SLAM, existing works often concentrate on specific components of the SLAM pipeline or emphasize algorithmic performance evaluated in structured or benchmarked environments. As a result, the broader evolution of visual and visual-inertial SLAM—particularly its deployment in unstructured natural environments and its transition toward learning-enhanced and semantic-aware systems—remains fragmented.

Special attention is given to environment-induced challenges—including dynamic vegetation, illumination variability, perceptual aliasing, uneven terrain, and degraded global positioning—and to how recent Deep Learning advances address these issues. Learned feature extraction, self-supervised depth estimation, semantic perception and end-to-end learning are discussed specifically in terms of their relevance and limitations for outdoor UGV deployments.

In this context, the present article provides a comprehensive review of the state of the art in VSLAM and VI-SLAM for UGV navigation in unstructured natural environments. The focus lies on three main axes:

- An analysis of visual and visual-inertial SLAM methods applied to unstructured natural environments, emphasizing how environment-induced factors such as dynamic vegetation, illumination changes, uneven terrain, and degraded global positioning affect system performance and robustness;
- A review of recent Deep Learning advances integrated into VSLAM and VI-SLAM pipelines, including learned feature extraction, depth estimation, semantic perception, and end-to-end learning, with a discussion of their benefits and limitations in outdoor deployments;
- Emerging opportunities for adaptive and multimodal sensor fusion strategies aimed at improving robustness, scalability, and long-term autonomy in agricultural and forestry applications.

Special attention is given to applications in the agricultural and forestry sectors, where robust and low-cost solutions are particularly relevant.

The structure of this article is as follows:

Figure 1 presents a diagram illustrating the structure of the article, divided by sections and their respective numbering.

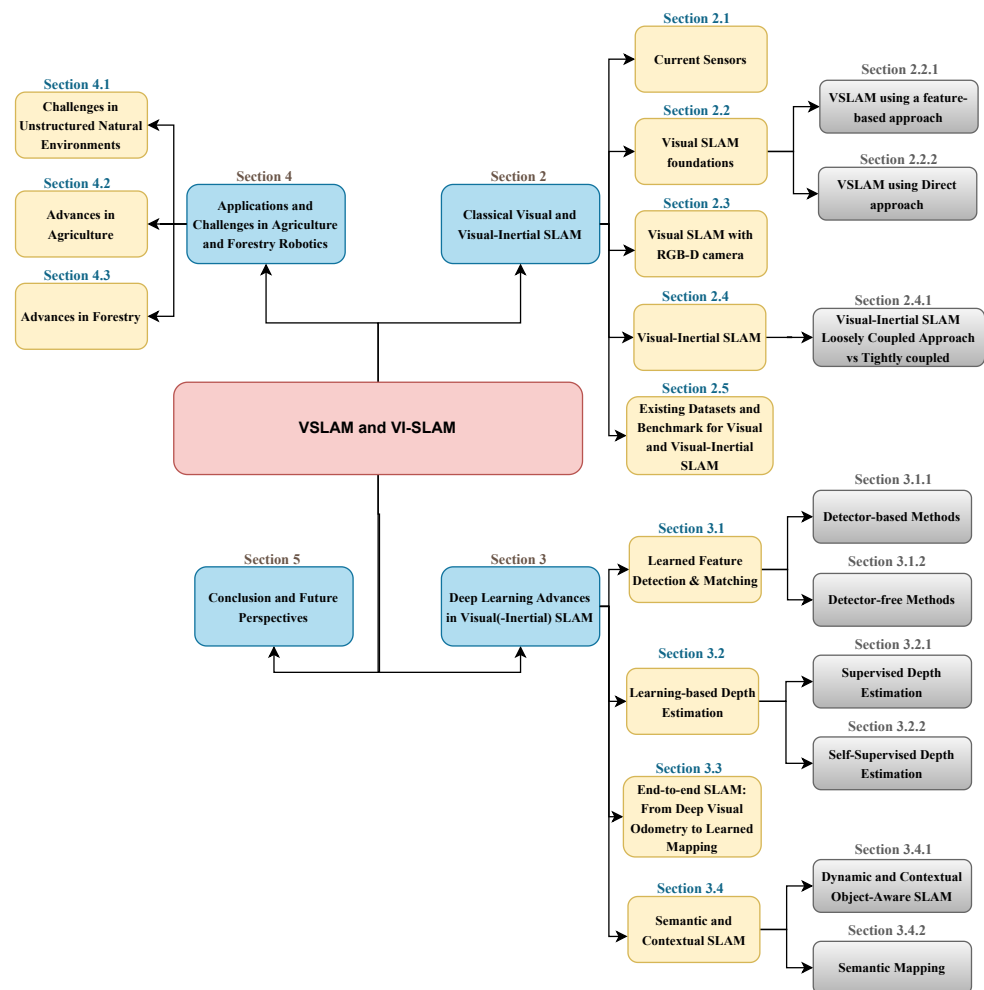


Figure 1. Diagram representing the structure of the article.

2. Classical Visual and Visual-Inertial SLAM

The purpose of SLAM is to build a representation of the environment surrounding a robot while simultaneously estimating its motion state within that environment. In the case of VSLAM, the technology has reached a level of maturity capable of achieving high accuracy on several datasets and in specific experimental evaluations. However, it still exhibits the following limitations [19]:

- **Illumination:** Difficulty handling low-light conditions or abrupt lighting changes, which negatively affect feature extraction and model optimization.
- **Motion:** Large or rapid camera movements may hinder algorithm convergence and result in tracking failure.
- **Dynamic scenes:** Moving objects are typically treated as outliers, impairing the correct modeling of the environment.
- **Noise:** Sensors—particularly low-cost ones—produce noisy data. Methods such as RANSAC [20] have limited effectiveness in highly noisy scenarios.

Thus, VSLAM technology still requires improvements in efficiency, robustness, and its ability to interpret dynamic scenes—limitations that are particularly relevant in unstructured natural environments.

2.1. Current Sensors

The sensors traditionally used in VSLAM are categorized into monocular cameras, stereo cameras, and RGB-D cameras. Both monocular and stereo systems operate based on similar principles, enabling their use in a variety of indoor and outdoor scenarios. RGB-D cameras, on the other hand, are a specific type of sensor capable of directly measuring scene depth, typically through active projection of structured infrared light or time-of-flight (TOF) techniques. Although easy to use, they are highly sensitive to lighting conditions and are generally restricted to indoor environments [21,22].

Accordingly, Figure 2 presents the characteristics of the different types of cameras traditionally used in VSLAM, as well as their respective advantages and disadvantages.

Camera Type	Operating Principle	Advantages	Disadvantages
 Monocular	Captures a single 2D image per frame, estimating depth from motion (<i>structure-from-motion</i>).	Low cost, reduced size and weight. Easy integration and low power consumption.	Does not measure depth directly and depends on motion.
 Stereo	Uses two cameras separated by a fixed distance (<i>baseline</i>) to estimate depth through image disparity.	Direct depth measurement. More robust to lighting variations and suitable for outdoor environments.	Requires precise calibration and slightly higher power consumption.
 RGB-D	Combines an RGB image with depth obtained via structured light or ToF.	Direct and accurate depth measurement, ideal for indoor scenarios. Simple integration.	Sensitive to ambient light, limited range and weak outdoor performance.

Figure 2. Main visual sensors used in VSLAM.

However, to complement VI-SLAM, another commonly fused sensor is the IMU. This sensor provides measurements of the robot's linear acceleration and angular velocity through accelerometers and gyroscopes, respectively. In some cases, these devices also incorporate magnetometers, which measure the Earth's magnetic field and provide additional information regarding absolute orientation [23].

These measurements enable the estimation of rapid pose changes between frames, serving as an additional source of information in situations where the camera becomes limited, such as environments with poor illumination, low texture, or the presence of occlusions. Despite these advantages, IMUs also present several limitations, most notably their susceptibility to Gaussian noise and drift caused by accumulated biases over time, which leads to integration errors [24]. For this reason, fusing IMU data with camera measurements is fundamental: the former offers high-frequency, environment-robust motion information, while the latter corrects accumulated drift using rich visual cues.

Table 1 presents the characteristics of the different sensors typically integrated in IMU units.

Table 1. Main components of the IMU sensor, with their respective measurements, advantages, and disadvantages.

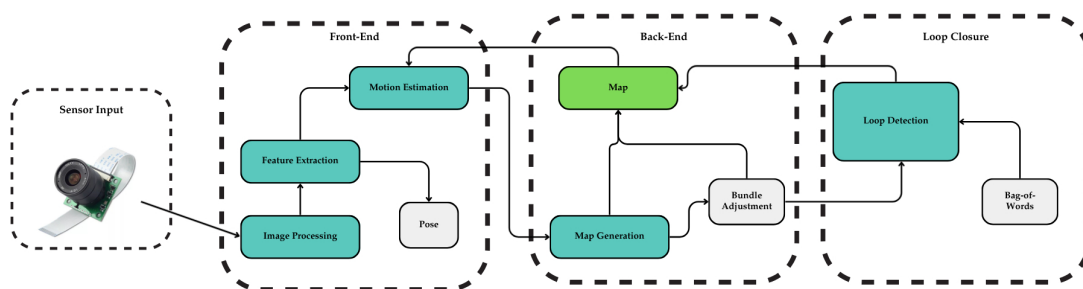
Component	Measurement	Advantages	Disadvantages
Accelerometer	Linear acceleration along the X, Y, and Z axes.	High sampling rate, fast motion detection, and low cost.	Susceptible to noise and drift when integrated.
Gyroscope	Angular velocity around the three axes.	Provides precise rotation measurements, essential for estimating orientation.	Suffers from cumulative drift over time.
Magnetometer	Intensity and direction of the Earth's magnetic field.	Allows absolute orientation reference, corrects gyroscope drift.	Sensitive to electromagnetic interference and variable accuracy depending on the environment.

2.2. Visual SLAM Foundations

A classical VSLAM system defines its pipeline as being divided into four main modules, as illustrated in Figure 3: the front-end tracking module, the back-end optimization module, loop closure detection, and map reconstruction. The front-end is responsible for estimating, in real time, the camera's position and orientation, thereby generating an odometry layer. The back-end receives the pose estimates produced by the front-end and refines them using optimization techniques—most commonly Bundle Adjustment (BA), among others—thus ensuring map updating and global consistency [25].

The map reconstruction module generates maps appropriate for the specific tasks to be performed, taking into account the camera trajectory across the captured frames. Finally, regarding Loop Closure Detection, there are situations in which the camera revisits the same locations. In such cases, it is necessary to identify previously visited places using methods such as bag-of-words or image-recognition-based techniques. The loop closure information is subsequently forwarded to the Pose Graph Optimization module [26].

Overall, advanced visual SLAM systems typically include modules for sensor data acquisition, front-end Visual Odometry, back-end optimization, loop closure detection, and map reconstruction.

**Figure 3.** VSLAM's General Architecture.

Regarding VSLAM architectures, they can be classified into three main categories [27]:

- Feature-based methods, which are generally robust but heavily dependent on the quality and stability of visual feature points.
- Direct methods, which minimize photometric error but are more sensitive to illumination changes.
- Semi-direct methods, which provide a balance between efficiency and robustness by combining characteristics of both feature-based and direct approaches.

2.2.1. VSLAM Using a Feature-Based Approach

To estimate the camera pose and construct a map of a given environment, Feature-Based VSLAM systems rely on feature detection, extraction, and matching [28]. This approach is among the most widely used, primarily due to its simplicity and practicality.

In the early stages of development, feature-extraction algorithms were largely based on corner detection techniques. One of the earliest examples is the Harris Corner Detector [29], which identifies corners by analyzing intensity variations under pixel shifts. More robust corner-based techniques were later introduced, such as the Shi–Tomasi Corner Detector [30] and FAST [31], among others. However, in scenarios involving image scaling, these earlier algorithms lose effectiveness, as a corner at one scale may no longer be identifiable at another. This limitation led to the development of more advanced feature-detection methods, beginning with SIFT [32], which extracts keypoints and computes associated descriptors. Subsequent techniques such as SURF [33], BRIEF [34], and ORB [35] were introduced, with ORB emerging as an efficient alternative to SIFT and SURF.

The Table 2 presents a comparison of traditional feature-extraction algorithms in terms of performance.

Table 2. Comparison of the main traditional feature extraction algorithms used in *Feature-Based VSLAM* methods, highlighting advantages and limitations.

Algorithm	Feature Type	Advantages	Disadvantages
Harris Corner Detector [29]	Corners (gradient-based)	Simple to implement	Sensitive to scale and rotation variations and does not generate descriptors.
Shi-Tomasi [29]	Corners (Harris improvement)	Higher stability and robustness than Harris and better selection of interest points.	Retains scale limitations and does not provide descriptors.
FAST [31]	Binary corners (circular segment)	Very fast and ideal for real-time applications.	Sensitive to noise and illumination variation and is not scale-invariant.
SIFT [32]	Multi-scale keypoints with gradient-based descriptors	Invariant to scale, rotation, and illumination, and high robustness.	High computational cost.
SURF [33]	Keypoints based on Haar wavelets	Faster than SIFT, good accuracy, and partially invariant to illumination.	Computationally heavy and sensitive to large perspective changes.
BRIEF [34]	Simple binary descriptor	Very efficient and easy to combine with fast detectors (e.g., FAST).	Not invariant to rotation or scale and low robustness in dynamic environments.
ORB [35]	Optimized combination of FAST + BRIEF	Fast, robust, rotation-invariant, and ideal for real-time applications (e.g., VSLAM).	Lower accuracy in complex textures.

After extracting the keypoints, the next step is feature matching between the detected points of interest in order to support subsequent pose estimation and optimization. The most basic method is Brute-Force Matching, which compares each keypoint descriptor from the current image with all descriptors from the reference image, evaluating similarity through descriptor distance. However, when the number of keypoints is high, this process becomes computationally expensive, potentially compromising real-time performance [36]. To address this issue, the FLANN (Fast Library for Approximate Nearest Neighbors) algorithm, introduced by Muja et al. [37], employs hierarchical K-means trees, significantly reducing search time [36,37].

After the feature correspondence stage, the relative camera pose is estimated using techniques such as epipolar geometry or the Perspective-n-Point (PnP) problem [38], typically in combination with RANSAC [20], to remove outliers. Depending on the type of sensor used in the visual SLAM system, three scenarios can be distinguished:

- Monocular camera: provides only 2D pixel coordinates; thus, motion estimation relies solely on epipolar geometry.
- Stereo or RGB-D camera: in addition to pixel coordinates, depth information is also available. In such cases, the literature often employs the Iterative Closest Point (ICP) algorithm [39] to estimate motion from two sets of 3D points.
- Configuration of 3D–2D : when a set of 3D points and corresponding 2D projections are available, motion estimation is performed using the PnP method [38].

In the back end, Bundle Adjustment (BA) is applied to jointly optimize the camera poses and the 3D point positions through a nonlinear optimization process [40], with frameworks such as g2o [41] and GTSAM [42] being widely adopted. Finally, loop closure detection identifies previously visited locations and corrects accumulated drift via global pose-graph optimization, thereby ensuring long-term map consistency [26].

This framework gave rise to several feature-based VSLAM pipelines, such as PTAM (Parallel Tracking and Mapping) [43]. In its front end, PTAM uses corner detection through the FAST algorithm to extract image features and subsequently estimate camera motion, while its back end performs nonlinear optimization and map construction. Another influential pipeline is ORB-SLAM [44], the first version to use the ORB algorithm for keypoint detection and tracking, enabling camera pose estimation, 3D map construction, and loop closure for global consistency. However, ORB-SLAM supports only monocular cameras, which motivated Mur-Artal and Tardós to develop ORB-SLAM2 [45], extending support to monocular, stereo, and RGB-D cameras. ORB-SLAM2 also achieved improved robustness in low-texture environments and scenes with limited depth. Subsequently, ORB-SLAM3 [46] was introduced (as illustrated in Figure 4), capable of operating either with pure visual input or through visual–inertial fusion incorporating IMU measurements. When running in purely visual mode, ORB-SLAM3 already provides enhanced accuracy, reduced drift, and better scalability compared to ORB-SLAM2. Another example is LineSLAM [47], which, instead of relying solely on keypoints, primarily uses line segments as geometric primitives to estimate camera pose and construct the environment map.

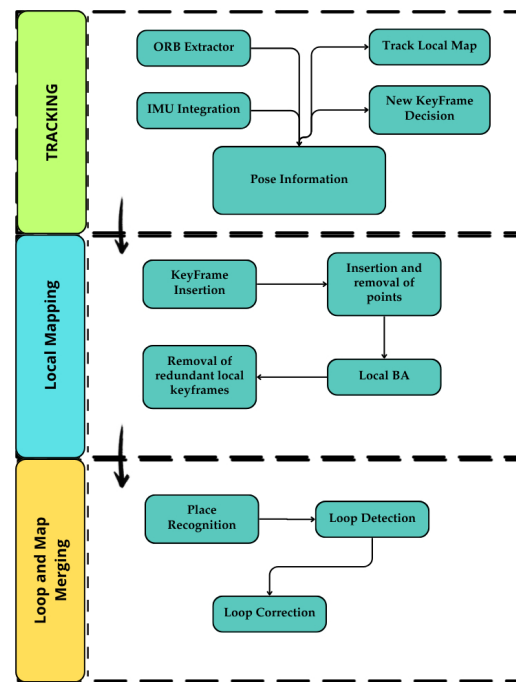


Figure 4. Flow chart of ORB-SLAM. Based on [46].

The Table 3 presents a comparison of the most widely used VSLAM pipelines.

Table 3. Comparison between the main feature-based pipelines in the VSLAM paradigm.

Pipeline	Features Used	Camera Type	Main Advantages	Limitations
PTAM [43]	FAST	Monocular	Simplicity and separation between tracking and mapping	Sensitive to illumination variations and reduced textures, and does not support loop closing
ORB-SLAM [44]	ORB	Monocular	High accuracy, loop closing detection, and 3D map construction	Compatible only with monocular cameras
ORB-SLAM2 [45]	ORB	Monocular, Stereo, and RGB-D	Greater robustness and support for different camera types	Accumulated drift over long trajectories
ORB-SLAM3 [46]	ORB (includes IMU support)	Monocular, Stereo, RGB-D	Better accuracy and stability, less drift, and better scalability	Higher complexity and computational cost
LineSLAM [47]	Lines	Monocular	Robust in environments with few textures and more structured geometric representation	Lower information density and sensitive to occlusions and illumination

2.2.2. VSLAM Using the Direct Approach

Compared to feature-based methods, the direct method operates immediately on pixel intensities, fully leveraging the information contained in the image. In other words, it bypasses the feature extraction and matching steps, resulting in higher computational efficiency. Furthermore, it uses sensor data without preprocessing, relying on the variation of pixel luminance and the minimization of photometric error to estimate motion and reconstruct the environment [48].

Several algorithms follow this methodology, and depending on the technique applied, the reconstruction can be dense, semi-dense, or sparse. However, reconstruction density represents a limiting factor for real-time operation, as the simultaneous optimization of structure and camera poses requires greater computational resources in dense and semi-dense reconstructions compared to sparse ones [48,49].

One of the first algorithms to emerge was proposed by Newcombe et al. [50], called DTAM, which uses all image pixels to construct an estimated depth map of a scene. However, this process incurs a high computational cost, both in storage and data processing, making real-time execution possible only with the use of a GPU. Consequently, LSD-SLAM disregards regions without texture to increase processing efficiency, allowing real-time execution using only a CPU. LSD-SLAM [51] is a relevant direct method because it integrates information extraction in featureless areas with semi-dense reconstruction, with a Visual Odometry core based on this reconstruction.

Forster et al. [52] proposed the SVO (Semi-Direct Visual SLAM) algorithm, which combines the benefits of feature-based methods with those of the direct method, organized into two main threads:

- Motion estimation, based on feature point matching;
- Mapping, performed by the direct method.

Although SVO achieves satisfactory results, being purely visual, it only performs short-term data association, which limits its accuracy [53]. Similarly, another algorithm is DSO (Direct Sparse Odometry) [54], which efficiently leverages all image pixels, maintaining robustness even in featureless areas, and obtaining more accurate estimates than SVO. It can determine the camera pose with high precision, even in low-texture or blurred image scenarios.

Table 4 presents a comparison of the different algorithms based on the direct method.

Table 4. Comparison between the main direct method-based algorithms.

Algorithm	Reconstruction Type	Main Advantages	Limitations
DTAM [50]	Dense	Detailed and accurate reconstruction and utilizes all the luminous information of the image Real-time execution with CPU only and good accuracy in environments with variable texture	High computational cost, requiring a GPU for real-time operation
LSD-SLAM [51]	Semi-Dense	Good efficiency and balance between accuracy and speed	Lower reconstruction density but sensitive to illumination changes
SVO [52]	Semi-Direct (Hybrid)	High accuracy even with little texture or blurry images	Short-term data association only, but with limited accuracy
DSO [54]	Sparse		Requires rigorous calibration and sensitive to illumination changes

2.3. Visual SLAM with an RGB-D Camera

RGB-D SLAM systems became popular starting in 2010 with Microsoft Kinect [55]. These sensors combine a monocular RGB camera with a depth sensor, allowing real-time depth maps to be obtained directly with affordable hardware. By providing the depth map directly, the architecture of RGB-D SLAM differs from traditional methods [11].

State-of-the-art research shows that most RGB-D systems use the Iterative Closest Point (ICP) algorithm to determine the sensor pose, combining depth maps to reconstruct the complete structure.

One of the first proposed systems is called KinectFusion [56], which enables real-time 3D reconstruction with an RGB-D camera. This system uses a point cloud derived from the depth information to calculate the camera pose using ICP. Subsequently, multiple point clouds captured across different frames are combined based on the estimated camera pose. However, the system does not include loop closure optimization, resulting in accumulated errors during prolonged operations, and the RGB information captured by the RGB-D camera is not fully exploited.

On the other hand, ElasticFusion [57] fully utilizes the color and depth data provided by the RGB-D camera. The camera pose is estimated by combining RGB color consistency with ICP, while accuracy is continuously improved through map optimization and reconstruction. To represent the map, it uses a surfel model, although its application is limited to small indoor scenes.

An enhanced version of KinectFusion is Kintinuous [58], which integrates loop closure by applying non-rigid transformations during rigid 3D reconstruction, allowing results from multiple loop closures to overlap and providing robust performance in indoor environments. Similarly, the RGB-D SLAM algorithm [59] is not only efficient but also comprehensive, integrating image feature modules, optimization, loop closure, and other functionalities, making it suitable for beginners to perform further development.

Table 5 presents a comparison of different RGB-D SLAM-based algorithms.

Table 5. Comparison of the main VSLAM algorithms that use RGB-D cameras.

Algorithm	Main Method	Map Representation	Main Advantages	Limitations
KinectFusion [56]	ICP	Point Cloud	Real-time 3D reconstruction, efficient use of depth maps, and high local accuracy	Absence of loop closing and accumulation of errors over time
ElasticFusion [57]	ICP + RGB Color Consistency	Surfel Model	Combines color and depth, continuous map optimization, and detailed, smooth reconstruction	Applicable only to small indoor scenes and high computational cost
Kintinuous [58]	ICP with Non-Rigid Transformations	Point Cloud	Integrates loop closing, supports continuous reconstruction of large environments, and provides robust performance	Higher computational complexity and dependent on precise calibration
RGB-D SLAM [59]	ICP + Visual Feature Extraction	Pose Graph	Modular and comprehensive system, includes optimization, loop closing, and integration of multiple modules	Lower accuracy compared to ElasticFusion and sensitive to depth noise

A major advantage of RGB-D systems is their ability to provide both color images and dense depth maps without the need for preprocessing, simplifying SLAM initialization.

However, they have some disadvantages regarding light sensitivity and exhibit limitations in narrow spaces, noisy environments, or areas with restricted fields of view, typically being confined to indoor applications [11].

2.4. Visual–Inertial SLAM

VSLAM-based algorithms have demonstrated robustness and progress; however, there are some situations that remain challenging, such as compensating for image blur caused by rapid camera movements or poor lighting conditions [60].

In this context, sensors such as IMUs can help improve the performance of VSLAM pipelines in certain scenarios, as illustrated in Figure 4, giving rise to VI-SLAM. VI-SLAM combines visual data from cameras with inertial measurements from an IMU to estimate the system's pose. Here, visual data is typically used for feature tracking and map construction, while the IMU provides motion estimates of a given UGV between frames. Nevertheless, VI-SLAM may sometimes introduce significant accumulated errors if visual tracking fails [61].

Currently, VI-SLAM is divided into two main approaches, as illustrated in Figure 5 [62]:

- Loosely coupled: In this approach, the IMU and camera independently estimate motion, and the respective poses are subsequently fused at a later stage.
- Tightly coupled: In this approach, data from the IMU and the camera at a given state are integrated into a single model, allowing for the joint formulation of motion and observation equations and, consequently, the simultaneous estimation of the system state.

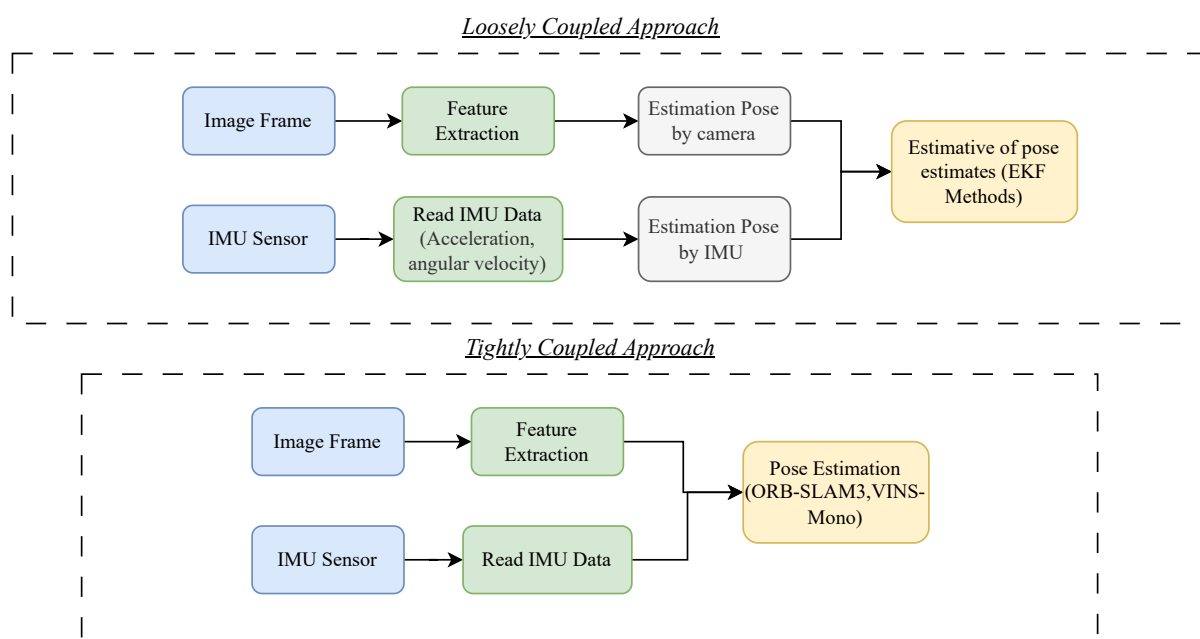


Figure 5. Difference between Loosely Coupled and Tightly Coupled VI-SLAM approaches

Thus, in the following subsections, some pipelines that illustrate the approaches described above will be discussed. State-of-the-art research shows that the most commonly used method is the tightly coupled approach, due to its higher robustness.

Visual–Inertial SLAM Loosely Coupled Approach vs. Tightly Coupled

The loosely coupled approach consists of fusing the positions and poses calculated by the visual sensor and the IMU. In this approach, most pipelines are based on filtering-based methods, with the most commonly used filter being the Extended Kalman Filter (EKF),

as shown in [63–65]. This method is simpler to implement than the tightly coupled approach but can be more susceptible to accumulated errors over long navigation periods [62].

The tightly coupled VI-SLAM approach involves integrating the states from the visual sensor and the IMU through an optimized filter. In this way, features extracted from images are incorporated into the state vector, allowing the simultaneous construction of motion and observation equations. Subsequently, the complete system state is estimated, yielding the pose information [66]. This type of coupling fully exploits visual and inertial information, which increases implementation complexity but provides superior pose estimation accuracy. Consequently, this method has become predominant in the field, with significant advances reported in its development.

One of the earliest research works on tightly coupled VI-SLAM was proposed by Mourikis et al. [67] and is called the Multi-State Constraint Kalman Filter (MSCKF). This algorithm performs tightly coupled fusion of visual and inertial data via EKF, demonstrating robustness to aggressive motion and texture loss while achieving good accuracy and real-time performance. It is widely used in robotics, Unmanned Aerial Vehicles (UAVs), and AR/VR, although the lack of loop closure limits the correction of accumulated errors. Another optimization-based algorithm is the Open Keyframe-based Visual-Inertial SLAM (OKVIS) [68], which fuses stereo vision with IMU data. This approach enables six-degree-of-freedom pose estimation. Although precise, it does not perform loop closure or construct complete global maps, which prevents it from being considered a full SLAM system. Similarly, in 2022, an enhanced version called OKVIS2 was proposed by Leutenegger et al. [69], introducing loop closure and global optimization, overcoming one of the main limitations of the original system that suffered from accumulated errors in long trajectories. Additionally, it implements an efficient marginalization and reactivation mechanism for observations, allowing the system to maintain real-time operation without losing accuracy. VINS-Mono [70], illustrated in Figure 6, was introduced in 2018 as a monocular algorithm with inertial fusion, supporting three integration modes: Monocular + IMU, Stereo + IMU, and Stereo only, with an optional GPS version. This system demonstrates high adaptability to outdoor environments and varying lighting conditions, though in navigation applications it is primarily used with UAVs. ORB-SLAM3 also supports the visual-inertial modality, as shown in Figure 5, enabling more robust and accurate motion estimation, particularly in dynamic and low-texture environments [46]. Recently, the IMU-aided Camera Extended Kalman Filter (ICEKF) [71] was proposed, which combines IMU and camera data and, through an adaptive weighting mechanism, adjusts the relative importance of visual and inertial measurements, thereby increasing robustness and adaptability in motion estimation.

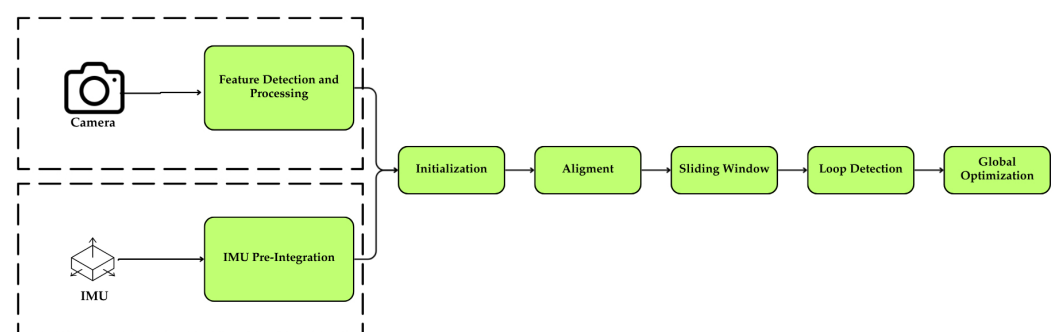


Figure 6. Overview of the VINS-Mono pipeline, where monocular visual feature processing and IMU pre-integration are tightly fused through initialization, alignment, and sliding-window optimization, with loop detection enabling global consistency. Based on [70].

In this way, the integration of inertial sensors in VSLAM has brought significant advantages, particularly in terms of the redundancy provided by multiple data sources. Thanks to the large amount of information obtained, the ease of use, and the low cost of visual sensors, the generated maps increasingly approximate human perception of the environment. Table 6 provides a comparison of the VI-SLAM algorithms using the Loosely Coupled and Tightly Coupled approaches discussed in this section.

Table 6. Comparison between the main VI-SLAM algorithms of the Loosely and Tightly Coupled approaches.

Algorithm	Coupling Type	Main Method	Limitations
EKF-based Methods [63–65]	Loosely Coupled	Extended Kalman Filter (EKF)	Prone to accumulated errors over long trajectories and lower global accuracy
MSCKF [67]	Tightly Coupled	Fusion via EKF (multi-state constraints)	Lacks loop closing and accumulation of errors in long operations
OKVIS [68]	Tightly Coupled	Keyframe-based Optimization	Absence of loop closing and global map, and accumulated drift over long trajectories
OKVIS2 [69]	Tightly Coupled	Global Optimization with Efficient Marginalization	More complex implementation and higher computational demand
VINS-Mono [70]	Tightly Coupled	Visual–Inertial Optimization (monocular, stereo, GPS)	Performance dependent on precise IMU and camera calibration
ORB-SLAM3 [46]	Tightly Coupled	Pose Graph-based Visual–Inertial Optimization	High complexity and greater consumption of computational resources
ICEKF [71]	Loosely Coupled	EKF Filter with Adaptive Weighting	Lower global accuracy compared to Tightly Coupled methods and dependent on calibration

2.5. Existing Datasets and Benchmark for Visual and Visual–Inertial SLAM

After the development of a SLAM algorithm or pipeline, its evaluation requires the introduction of tools that allow performance assessment. This evaluation essentially depends on the availability of datasets, which enable the verification of the algorithm's performance not only at the sensor level but also under various environmental conditions. The standardization of benchmarks has been fundamental for the advancement of the field, allowing an objective analysis of factors such as accuracy, robustness, execution time, and consistency of the estimated trajectory. However, most available datasets were

designed in urban or controlled environments, presenting a limited degree of complexity in terms of terrain, lighting, and environmental dynamics—factors that differ significantly in unstructured natural environments where UGVs operate.

One of the most famous and traditional datasets was proposed by Geiger et al. [72] in 2013, called the KITTI Vision Benchmark Suite. It is widely used due to its combination of stereo, LiDAR, and GNSS/RTK data, covering urban and suburban trajectories. However, its flat terrain and controlled lighting conditions limit its applicability to agroforestry scenarios or dense vegetation environments. In 2016, the EuRoC MAV dataset [73] was introduced, designed for drones in industrial indoor environments, providing precise calibration and synchronization between IMU and stereo cameras, but it does not represent outdoor navigation challenges. The TUM VI dataset [74], in turn, introduces trajectories in partially illuminated environments and combines high-frequency sensors, being particularly useful for visual–inertial calibration tests, though still predominantly indoor. These datasets (KITTI, EuRoC MAV, and TUM VI) have served as standard references for evaluating SLAM systems, including VSLAM and VI-SLAM.

In recent years, datasets targeting complex, unstructured outdoor environments have emerged, driven by the growing need to validate SLAM algorithms under real-world field conditions. Among these, datasets related to agricultural and forestry scenarios are particularly relevant.

In agricultural contexts, notable examples include the Rosario Dataset [75], developed by Cortés et al., collected in agricultural fields. It includes stereo, IMU, and GNSS/RTK data, representing realistic contexts of uneven terrain and variable lighting. Figure 7 shows an example extracted from this dataset. In 2023, Polvara R. et al. developed the Bacchus Long-Term (BLT) dataset [76], which involved deploying an autonomous mobile robot in a vineyard to perform multimodal data collection sessions, resulting in the creation of the BLT dataset. This dataset also explores stable temporal features to improve 4D localization. More recently, the MAgro dataset [77] was introduced, consisting of agricultural scenarios where data were collected in apple and pear orchards, repeating the same trajectories on different days under varying light and weather conditions to represent real-world variability. The dataset was collected using a Clearpath Jackal UGV equipped with frontal and lateral stereo cameras, a 3D LiDAR, three IMUs, and wheel encoders. The dataset also includes calibrated RTK-GPS, which serves as ground truth for evaluating algorithm accuracy.

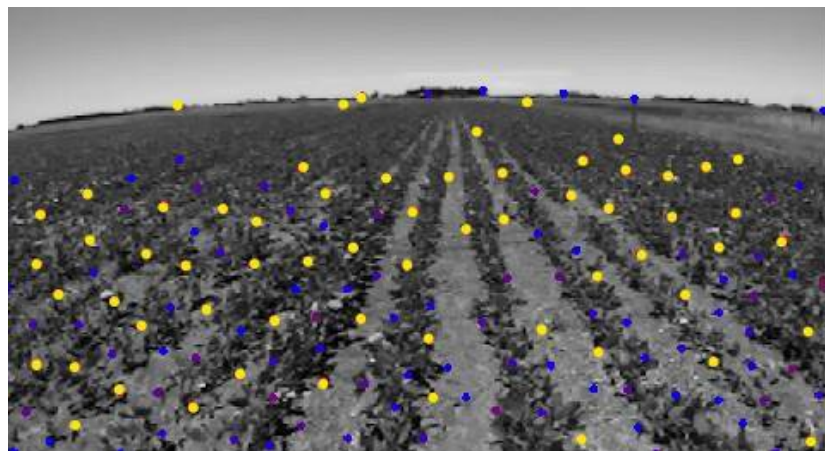


Figure 7. Example of data collection from the ROSARIO dataset (reprinted from ref. [75], licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>) (accessed on 1 December 2025))).

Regarding forested environments, the RELLIS-3D Dataset [78] stands out. This is a multimodal dataset collected in off-road environments to support the development of 3D

semantic segmentation in autonomous robotics. The dataset includes 13,556 LiDAR scans and 6235 images, with detailed annotations and challenges such as class imbalance and irregular terrain. Figure 8 shows an example of a data capture from this dataset. Similarly, the FinnForest Dataset [79] was specifically designed for forest environments, offering multiple trajectories under different seasonal conditions (summer and winter), making it particularly valuable for studies on robustness and variation in visual textures. Orty T. et al. developed the GROUNDED dataset [80], created to improve the localization of robots and autonomous vehicles under variable environmental conditions, albeit less affected by weather or surface changes. This dataset includes approximately 108 routes, totaling 450 km of driving in diverse weather conditions and environments, with synchronized data from LiDAR, GPS RTK/INS, and cameras.

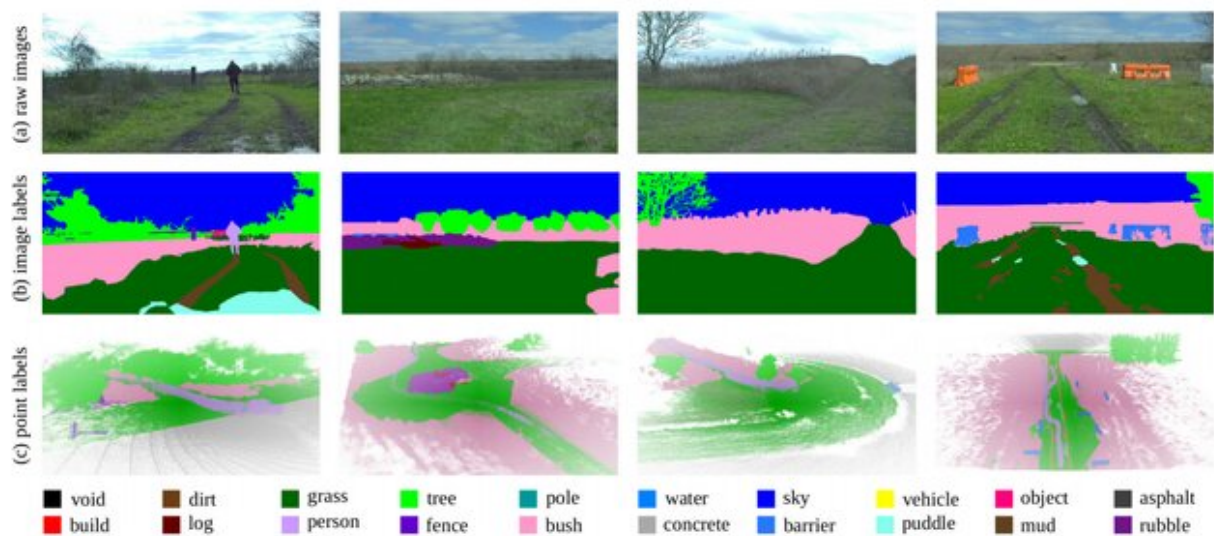


Figure 8. Example of data collection from the RELLIS-3D Dataset (reprinted from ref. [78], licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>) (accessed on 1 December 2025))).

Finally, some datasets attempt to combine multiple scenarios, such as the Rover Dataset [81] and the M2UD Dataset [82], which have expanded this domain with multi-modal collections in agricultural and rural environments. These datasets include RGB-D cameras, IMU, GNSS, and irregular terrain data, providing new challenges for sensor fusion and visual matching algorithms.

Despite recent advances, there is still a lack of suitable datasets for Visual-Inertial SLAM in unstructured natural environments. Existing datasets generally do not provide precise synchronization between sensors or present inaccurate ground truth due to GNSS failures under dense vegetation. Additionally, some do not include seasonal, weather, or lighting variations, which makes evaluating the generalization of Deep Learning models difficult. This scarcity of data limits the comparison and validation of modern pipelines, particularly those employing self-supervised learning or attention mechanisms.

Table 7 presents a comparison of the main datasets used in SLAM and VI-SLAM research, considering factors such as available sensors, environment type, presence of accurate the ground truth, and level of environmental complexity.

Table 7. Comparison of the main *datasets* used in VSLAM and VI-SLAM.

Dataset	Main Sensors	Environment Type
KITTI [72]	Stereo, 3D LiDAR, IMU, GNSS/RTK	Urban/Suburban
EuRoC MAV [73]	Stereo cameras, IMU	Indoor (industrial scenarios)
TUM VI [74]	Stereo, IMU, ambient light sens.	Indoor/Semi-outdoor
Rosario [75]	Stereo, IMU, GNSS/RTK	Agricultural (fields)
BLT (Bacchus Long-Term) [76]	Stereo, LiDAR, IMU, GNSS	Vineyards
MAgro [77]	Front/side Stereo, LiDAR, multiple IMUs, encoders, GPS RTK	Orchards (apple, pear)
RELLIS-3D [78]	3D LiDAR, RGB cameras, GNSS/INS	Off-road/Natural
FinnForest [79]	RGB, IMU, GNSS	Forest (summer/winter)
GROUNDED [80]	Cameras, LiDAR, GPS RTK/INS, LGPR	Mixed (rural/urban)
Rover Dataset [81]	RGB-D, IMU, GNSS	Rural/Natural environments
M2UD [82]	RGB-D, LiDAR, IMU, GNSS/INS	Urban/Rural/Off-road

3. Deep Learning Advances in Visual(–Inertial) SLAM

In recent years, advances in Deep Learning have revolutionized the way SLAM systems perceive and interpret the environment. Unlike classical methods, which rely on manual heuristic approaches and geometric descriptors, Deep-Learning-based approaches enable the extraction of more robust representations. These improvements have led to the emergence of end-to-end pipelines capable of learning the complete mapping and localization process.

Deep Learning has been applied primarily in four key areas of the VSLAM pipeline:

- Motion estimation: Deep neural networks are used to learn pose transformations, sometimes eliminating the need for feature correspondence.
- Global relocalization: Models learn more robust spatial representations capable of recognizing previously visited locations, reducing accumulated drift.
- Mapping: Neural networks extract both the geometry and semantics of the scene, enabling the creation of richer and more interpretable maps.

According to recent reviews [83,84], Deep Learning approaches applied to SLAM can be organized into four categories:

- Modular learning: Neural networks replace specific components of the classical pipeline, such as feature detection, depth estimation, or sensor fusion.
- Joint learning: Multiple modules are trained jointly, approaching fully integrated (end-to-end) pipelines.
- Confidence learning: Introduces dynamic weighting mechanisms and uncertainty estimates to improve the robustness of sensor fusion.
- Active learning: Explores adaptive strategies for intelligent selection of views, trajectories, or sensors during the mapping process.

Within this framework, modular and joint learning approaches have been the most widely explored in VSLAM and VI-SLAM, showing a promising balance between performance, interpretability, and robustness. In modular approaches, neural networks are integrated into essential and specific components of classical pipelines, replacing deterministic algorithms with neural modules, for example, in feature detection and matching and/or depth estimation from monocular cameras. On the other hand, joint learning approaches aim for deeper integration between modules, training perception, mapping, and optimiza-

tion stages together, thus generating end-to-end systems that directly learn both the motion dynamics of the environment and its 3D reconstruction (End-to-End SLAM).

Additionally, Deep Learning has driven emerging areas such as Semantic and Contextual SLAM, which assigns meaning and context to visual observations, and Dynamic Object-Aware SLAM, which combines semantic perception and motion detection to distinguish between dynamic and static elements in a given environment. These aspects are crucial for ensuring robustness in complex natural contexts, such as agricultural and forest environments.

These developments demonstrate that Deep Learning is not merely a substitute for classical components but rather a complementary paradigm that redefines the balance between geometry, semantics, and adaptability in modern UGV SLAM systems.

3.1. Learned Feature Detection and Matching

The matching of local features in a given scene has proven to be a crucial and challenging task in computer vision applications, attracting significant attention from researchers.

In the context of traditional VSLAM, feature matching typically follows a detection-based approach, in which keypoint detectors are used to extract and describe image features. However, with the advances in Deep Learning, new methods have emerged that demonstrate improved performance and computational efficiency. According to [85,86], the existing image matching methods can be broadly classified into two categories: detection-based methods and detector-free methods.

3.1.1. Detector-Based Methods

The detector-based methodology follows the classical VSLAM pipeline. However, when combined with Deep Learning, the approach replaces traditional methods with neural networks capable of learning keypoints and descriptors directly from visual data.

Detector-based methods can be further classified into four categories:

- **Detect-then-Describe:** In this method, sparse point correspondences are used. It follows the “detect and then describe” paradigm: first, keypoints are identified, and then descriptors are extracted from small regions around them. Training employs metric learning, aiming for similar points to be close and dissimilar points to be far apart in feature space. For greater efficiency, detectors focus on small regions of the image [86].
- **Joint Detection and Description:** Although sparse local feature matching is effective, it is limited under environmental changes. Therefore, Joint Detection and Description integrates keypoint detection and descriptor extraction within a single model, learning both simultaneously. This allows the model to combine detection and description information, better adapting to varying image conditions and capturing higher-level features such as shape and structure. Compared to traditional methods that train detectors and descriptors separately, this approach reduces information loss and inconsistencies, producing richer descriptors and improving feature matching, especially in challenging scenarios like illumination changes or weak textures [87].
- **Describe-then-Detect:** This method is used to extract local image features by first describing image regions using information-rich descriptors and then detecting keypoints from these descriptions. Instead of detecting points first and then extracting descriptors, this pipeline focuses on the information contained in the descriptors to identify the most relevant points [88].
- **Graph-based Methods:** In traditional feature matching, keypoints are paired using nearest-neighbor search of the descriptors, and outliers are removed based on scores or mutual verification. More recently, attention-based Graph Neural Networks (GNNs) have been employed, treating keypoints as nodes and using self-attention and cross-

attention mechanisms to exchange visual and geometric information between them. This enables more accurate correspondences, overcoming the limitations of isolated local descriptors through a soft assignment matrix [89].

In this way, several detector-based Deep Learning methods have emerged to address some of the limitations of traditional approaches. One of the earliest methods, introduced in 2016, is LIFT (Learned Invariant Feature Transform) [90], whose primary goal is to automatically learn the entire feature extraction process using deep neural networks. It integrates three components: a keypoint detection network, an orientation estimation network, and a descriptor network, all trained jointly. In 2018, DeTone et al. proposed SuperPoint [13], which has become one of the most widely used methods for local feature matching with Deep Learning. Its operation is based on a convolutional neural network that takes an image as input and produces two output maps: a keypoint map, indicating the most relevant regions of the image, and a descriptor map, providing a unique vector representation for each detected point. Subsequently, in 2020, SuperGlue [91] was introduced. Instead of relying on heuristics (such as Euclidean distance) to compare descriptors, SuperGlue learns contextual relationships between features in the two images, effectively “learning to match points that truly make sense together.” Inspired by SuperGlue, LightGlue [14], as illustrated in the architecture diagram in Figure 9, was designed for efficiency and real-time applications. It uses adaptive attention and avoids redundant computations for easy images, i.e., images with few matches. Finally, Alike (Accurate and Lightweight Keypoint Detection and Descriptor Extraction) [92] is a Deep-Learning-based method developed to perform keypoint detection and description simultaneously, with both precision and efficiency. It introduces a differentiable approach that allows joint optimization of keypoint locations and their descriptors through backpropagation. This method employs mechanisms such as local soft-argmax, enabling sub-pixel-level detection and description, thereby improving spatial accuracy.

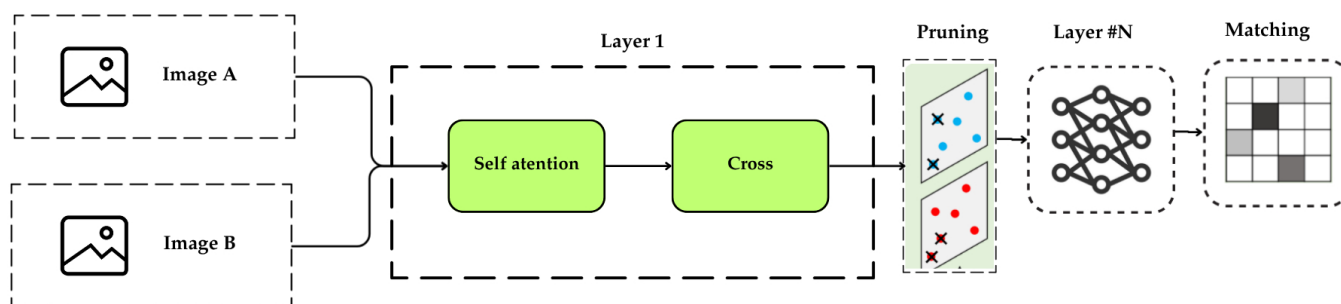


Figure 9. Overview of the LightGlue pipeline, where features from two images are processed through self-attention and cross-attention layers, followed by adaptive pruning and final matching to efficiently establish reliable feature correspondences [14].

Consequently, Table 8 presents a comparison of the different detector-based Deep Learning methods and their respective attributes.

3.1.2. Detector-Free Methods

The feature detection stage helps reduce the search space for matching but becomes challenging in extreme situations, such as images with large viewpoint changes or textureless regions, even when using highly accurate descriptors and matching methods [93]. In contrast, detector-free methods eliminate the keypoint detection step and extract visual descriptors directly on a dense grid across the image, enabling the generation of dense correspondences. Compared to detector-based methods, these approaches can identify keypoints with higher repeatability across different images.

Table 8. Comparison between detector-based Deep Learning methods and their main attributes.

Method	Year	Main Characteristics	Advantages and Limitations
LIFT [90]	2016	Jointly learns the steps of detection, orientation estimation, and keypoint description using deep neural networks.	Allows end-to-end learning, eliminating manual steps; however, it has a high computational cost and lower real-time speed.
SuperPoint [13]	2018	CNN that generates two maps: one for keypoints and one for descriptors. Trained using a self-supervised approach.	A balance between accuracy and performance, and widely used in local feature matching and SLAM tasks. May degrade in low-texture environments.
SuperGlue [91]	2020	Uses attention and graph neural networks to learn contextual relationships between keypoints from two images.	Provides highly robust matching and outperforms traditional heuristics. However, it is computationally demanding.
LightGlue [14]	2023	Optimized version of SuperGlue with adaptive attention and dynamic computation based on the matching difficulty.	High efficiency and suitable for real-time applications. Slight loss of accuracy in complex scenarios.
ALIKE [92]	2021	A lightweight and accurate method that integrates differentiable detection and description via backpropagation, uses local soft-argmax for sub-pixel accuracy.	Excellent compromise between accuracy and efficiency, and ideal for robotics and SLAM. Sensitive to environments with low texture.

In the detector-free methodology, the first approaches were CNN-based, using correlation or cost volumes to recognize possible local consistencies between image neighborhoods [86]. One of the earliest methods in this domain, introduced in 2020, was NCNet [94], which bypasses the initial step of identifying interest points and instead considers all possible correspondences between pixels of the two images. DualRC-Net [95] proposed an innovative approach for determining dense correspondences between image pairs through a coarse-to-fine strategy. Based on a dual-resolution architecture inspired by the Feature Pyramid Network (FPN), the method constructs a 4D correlation tensor from low-resolution feature maps and subsequently applies a learnable neighborhood consensus module. This combination progressively refines correspondences, enhancing both robustness and accuracy in locating corresponding points. Building upon these advances, RAFT (Recurrent All-Pairs Field Transforms) [96] significantly improves dense correspondence estimation by extracting per-pixel features and constructing multi-scale 4D correlation volumes for all pixel pairs. This method introduces a recurrent processing unit that iteratively refines the flow field, allowing continuous and precise updates. This innovative approach overcomes limitations of previous methods, such as error propagation at coarse resolutions and difficulties handling small or fast-moving objects, resulting in a more accurate, stable, and reliable flow estimation. Deep Feature Matching (DFM) [97] relies on a pre-trained Visual Geometry Group (VGG) network as a feature extractor, efficiently capturing correspondences without requiring additional training. This approach demonstrates the robustness and generalization capability of deep-layer features from VGG, which is widely recognized for its effectiveness in computer vision tasks. A more recent method, ConDL (Contrastive Descriptor Learning) [98], proposed by Kwiatkowski et al. (2024), is a CNN-based detector-free dense image matching framework. As illustrated in Figure 10, ConDL employs contrastive learning to pull corresponding descriptors closer and push non-corresponding ones apart, generating dense per-pixel feature maps. Trained with syn-

thetic data and various distortions, ConDL exhibits robustness to viewpoint and lighting changes, while its differentiable sampling reduces computational cost.

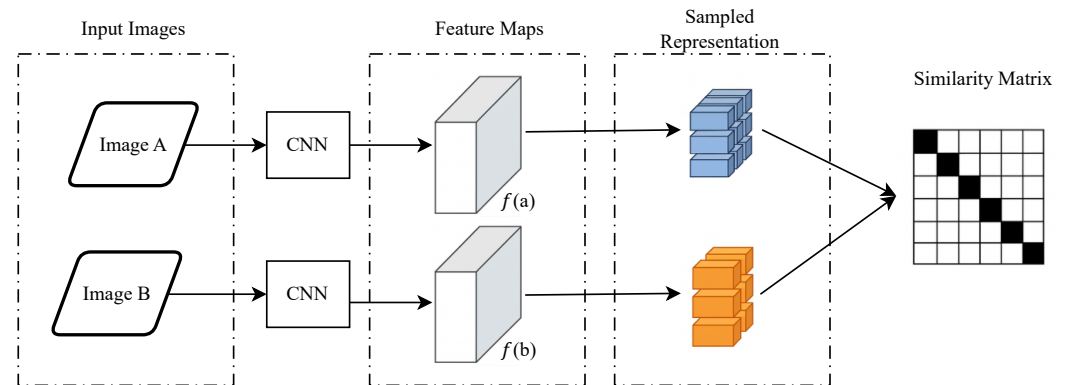


Figure 10. Diagram of the Detector-Free ConDL method architecture based on [98].

Despite advances in the field of CNN-based dense methods, these networks face limitations in handling textureless regions and distinguishing keypoints with similar characteristics. This has led to the emergence of Transformers, which are models that use attention mechanisms to process sequences of data, allowing each element to consider relationships with all others without relying on sequential processing [99]. This type of model has demonstrated significant advantages in computer vision and has consequently attracted interest from researchers for applications in feature matching.

Several approaches have emerged in this subfield of computer vision. For instance, COTR (Correspondence Transformer) [100] integrates the benefits of both sparse and dense matching, learning correspondences between two image pairs simultaneously. It uses self-attention with selected keypoints as queries and recursively refines correspondences in the second image through a neural network, unifying the entire process into a parameter-optimized problem. Another highly relevant and innovative method is LoFTR (Detector-Free Local Feature Matching with Transformers) [101], which employs a Graph Neural Network (GNN), treating keypoints as nodes and combining self-attention and mutual attention to generate dense descriptors. An extension of LoFTR, SE2-LoFTR [102], incorporates a steerable CNN, achieving invariance to translations and rotations, which significantly enhances the model's robustness to rotational movements and complex image scenarios. These models demonstrate high robustness in low-texture regions and efficiently integrate local and global information. Wang Q. et al. [103] proposed MatchFormer, a model that uses a hierarchical encoder with self-attention and cross-attention to capture both local and global relationships, improving accuracy in low-texture areas or repetitive patterns. One of the most recent and promising approaches is MatchAnything, proposed by He, X. et al. [104]. This model generates dense, generalizable descriptors capable of finding correspondences between images with significant differences in appearance, scale, or modality. It represents an evolution in feature matching, expanding the concept of dense correspondence to multimodal and highly generalizable applications. By training on a wide variety of synthetic cross-modal data, the model learns more fundamental structural correspondences that are less dependent on pointwise appearance. Its applications range from autonomous systems to computer vision and medical imaging, as illustrated in Figure 11.

In summary, detector-free methods currently represent the most advanced approach for local and dense feature matching, particularly in natural and unstructured environments where texture, lighting, perspective, and shape variability are high. By dispensing with keypoint detection, these methods can generate dense and repeatable descriptors across the entire image, efficiently integrating local and global information. Models such as COTR, LoFTR/SE2-LoFTR, MatchFormer, and MatchAnything demonstrate robustness in

complex scenarios, textureless regions, and images with significant variations, including multimodal differences. Compared to detector-based methods, detector-free approaches offer higher accuracy, generalization, and reliability, making them the ideal choice for feature matching applications in dynamic and challenging natural environments, where traditional methods tend to fail. The Table 9 provides a comparison of the various methods presented in this subsection.

Table 9. Comparison between detector-free methods and their main attributes.

Method	Network Type	Main Characteristics	Application/Highlight
NCNet [94]	CNN	Considers all possible pixel-to-pixel correspondences through a 4D correlation volume, eliminating keypoint detection.	First widely used method in the <i>detector-free</i> category and a strong basis for subsequent methods.
DualRC-Net [95]	CNN + FPN	Coarse-to-fine strategy with 4D volume and learnable neighborhood consensus, improving accuracy in complex regions.	High robustness and accuracy in dense correspondence.
RAFT [96]	CNN	Utilizes multi-lateral 4D volumes and a recurrent unit for iterative refinement, providing highly accurate optical flow.	Excellent performance in optical flow and widely adopted as a baseline.
DFM [97]	Pre-trained VGG	Deep feature extractor without the need for additional training, focused on robustness and generalization.	Suitable for scenarios with severe changes in illumination and appearance.
ConDL [98]	CNN	Contrastive learning with dense per-pixel maps and differentiable sampling, improving geometric invariance.	Robust to large variations in perspective and illumination.
COTR [100]	Transformer	Integrates self-attention mechanisms and dense/sparse matching with recursive refinement.	Combines advantages of dense and keypoint-based approaches.
LoFTR [101]	Transformer + GNN	Models local and global dependencies via self-attention and cross-attention, completely replacing keypoint detection.	Highly effective in areas with low texture and homogeneous surfaces.
SE2-LoFTR [102]	Transformer + Steerable CNN	Rotation and translation equivariant architecture; suitable for scenarios with strong geometric variations.	Higher robustness to rotational movements and spatial changes.
MatchFormer [103]	Hierarchical Transformer	Hierarchical model with multi-scale local and global relationships for dense correspondence.	Very accurate in environments with repetitive patterns.
MatchAnything [104]	Transformer	Universal cross-modal model trained on a large scale and supports large differences in appearance, scale, and modality.	Multimodal correspondence (RGB–IR, RGB–Depth, satellite, etc.).

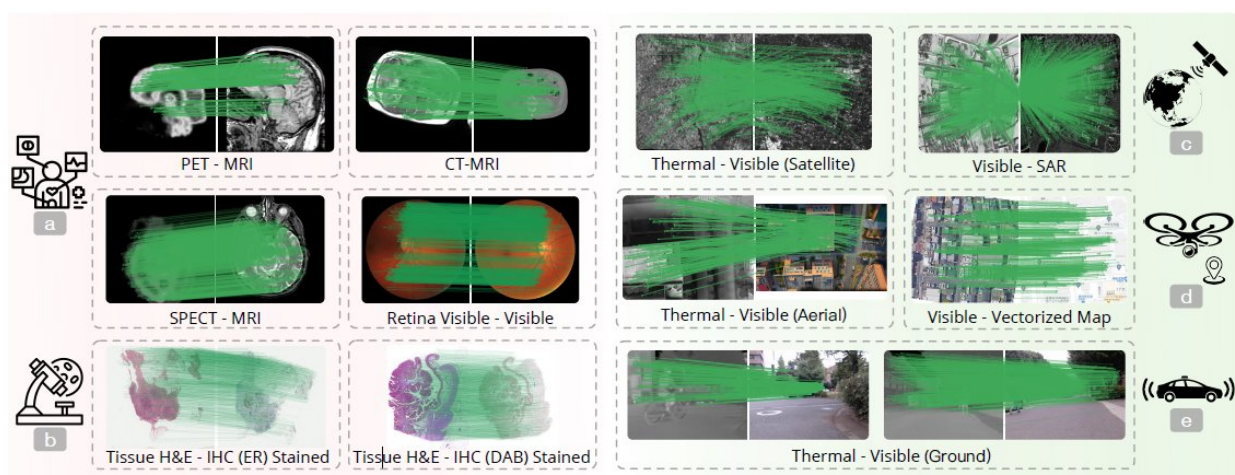


Figure 11. Applications of the MatchAnything framework. The green lines indicate the location of the corresponding pixels between images (reprinted from ref. [104], licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/> (accessed on 1 December 2025))).

3.2. Learning-Based Depth Estimation

Depth estimation is a fundamental component for both VSLAM and VI-SLAM, as it enables the reconstruction of the three-dimensional structure of an environment and, in turn, allows for estimating the absolute scale of the trajectory. In classical methods, depth is obtained through epipolar geometry (using stereo or multiview configurations); however, in monocular setups, this information becomes ambiguous, leading to accumulated errors.

The emergence of Deep Learning has opened opportunities for new approaches by introducing neural networks capable of estimating depth maps directly from RGB images, reducing dependence on geometric calibrations and increasing robustness in environments with occlusions, variations in texture, and lighting conditions—common in unstructured natural environments. Approaches that integrate Deep Learning can be grouped into two main categories:

1. Supervised, which require ground-truth depth data.
2. Self-supervised, which learn from photometric consistency between consecutive images.

3.2.1. Supervised Depth Estimation

Depth estimation from a single image is inherently an ambiguous problem, as the same two-dimensional projection can correspond to multiple possible three-dimensional configurations. To overcome this limitation, Deep-Learning-based methods have demonstrated high effectiveness by modeling the relationship between RGB images and their corresponding depth maps, enabling more consistent and realistic predictions of scene geometry.

In this approach, methods attempt to infer the scale approximately based on images from a specific dataset [105]. Networks are thus trained using ground-truth depth.

One of the earliest seminal works is by Eigen et al. (2014) [106], which introduced a multi-scale convolutional architecture capable of predicting depth and semantic labels from a single RGB image. Later, in 2016, Laina et al. [107] proposed a fully convolutional network with residual blocks (ResNet), which substantially improved the accuracy and density of depth predictions. Finally, DeepV2D [108] is an end-to-end architecture for monocular depth estimation in videos, combining two modules: one that generates an initial depth map based on camera motion, and another that refines camera motion using the predicted depth.

Despite excellent results in controlled environments, these supervised approaches rely heavily on precise depth sensors and face challenges in generalizing to natural outdoor environments, where lighting, texture, and vegetation vary significantly.

3.2.2. Self-Supervised Depth Estimation

To mitigate the limitations of supervised approaches, self-supervised methods emerged to avoid the high cost and difficulty of collecting ground-truth depth using sensors such as LiDAR. These methods enable large-scale training with inexpensive data, which is crucial for 3D reconstruction applications. This approach learns to predict depth at the pixel level using monocular videos without depth labels, training two networks (depth and pose) to reconstruct the target frame from neighboring frames via reprojection geometry and photometric error minimization, including masks for static pixels, edge-guided smoothness regularization, and multi-scale losses [109].

One of the first pioneering works in this field was proposed in 2017 by Zhou T. et al. [110] with SfMLearner, which employs two coupled networks: one to estimate depth and another for camera pose, trained using photometric loss. However, SfMLearner suffers from scale ambiguity, as there is no external reference for size or distance. Subsequently, to improve parameters such as scale consistency, MonoDepth2 [15] was introduced. MonoDepth2 performs self-supervised training with a depth network and a pose network, using photometric

reconstruction through frame reprojection and minimum per-pixel reprojection loss to handle occlusions. It has since become the classic baseline for subsequent methods. Following this, HR-Depth [111] was proposed to overcome limitations of earlier methods such as MonoDepth2. It redesigns skip connections in DepthNet to better preserve high-resolution spatial and semantic information and incorporates a feature fusion Squeeze-and-Excitation (fSE) module to improve depth representation.

More recently, transformer-based architectures and self-attention mechanisms have shown significant gains in generalization and accuracy, making their integration into SLAM pipelines increasingly promising. For example, Yan J. et al. (2021) proposed CAdDepth [112], structured around two main modules: a structure perception module, which applies self-attention to integrate long-range information and extract rich scene representations, and a detail enhancement module, which recalibrates feature maps to highlight important local information. These modules work together to produce more precise and sharper depth predictions. SwinDepth [113] employs Swin Transformers, leveraging hierarchical attention and sliding windows to capture long-range dependencies and multi-scale spatial relationships, efficiently integrating global and local information to improve accuracy in complex or low-texture regions. Tang S. et al. proposed CATNet (Convolutional Attention and Transformer for Monocular Depth Estimation) [114], using a simple encoder–decoder architecture coupled with attention modules, including MCA (Multidimensional Convolutional Attention) to refine multi-scale features and DAT (Dual Attention Transformer) to capture global relationships, as illustrated in Figure 12. This combination enables CATNet to produce accurate depth estimates with reduced computational complexity. More recently, Liu et al. [115] proposed a transformer-based monocular depth estimation model combining hybrid attention and progressive regression to improve depth map accuracy and consistency. The method includes a hybrid attention fusion module, combining the benefits of convolutions (local detail capture) and transformers (global dependency modeling), for richer feature extraction. Additionally, it integrates a deep progressive regression module that predicts depth maps at different resolutions, gradually refining details while preserving spatial structure. Finally, PAR-mono [116] is a dynamic attention and channel separation-based approach to improve depth prediction accuracy and temporal consistency. The method uses depth estimated from the previous frame to reinforce temporal continuity and employs an adaptive encoder with variable convolution to preserve spatial coherence.

Table 10 presents an analysis of the various methods discussed regarding depth estimation with Deep Learning:

Table 10. Comparison between different Deep-Learning-based depth estimation methods.

Method/Reference	Year	Supervision Type	Architecture/Base	Main Contributions	Limitations/Disadvantages
Eigen et al. [106]	2014	Supervised	CNN multi-scale	First network to estimate depth, normals, and semantics from an RGB image.	Low resolution and dependence on labeled data.
Laina et al. (ResNet) [107]	2016	Supervised	FCN + ResNet	Introduction of residual blocks, improving the density and accuracy of estimates.	Poor generalization in outdoor environments.
DeepV2D [108]	2019	Supervised	Dual Network (Depth + Motion)	Integrates depth and motion in videos into an end-to-end model.	Requires precise calibration.

Table 10. Cont.

Method/Reference	Year	Supervision Type	Architecture/Base	Main Contributions	Limitations/Disadvantages
SfMLearner [110]	2017	Self-supervised	Dual CNN (Depth + Pose)	Introduced photometric reconstruction-based training.	Scale ambiguity and sensitivity to dynamic objects.
MonoDepth2 [15]	2019	Self-supervised	CNN + PoseNet	Improved photometric reprojection and scale consistency.	Limited in poor textures and dynamic scenes.
HR-Depth [111]	2021	Self-supervised	Redesigned DepthNet + fSE	Preserves semantic and spatial details and optimized connections.	Higher computational cost.
CADepth [112]	2021	Self-supervised	Transformer + Detail Emphasis	Integrates global self-attention and emphasizes local details.	High computational cost.
SwinDepth [113]	2022	Self-supervised	Swin Transformer	Long-range dependencies with hierarchical windows.	High memory usage.
CATNet [114]	2023	Self-supervised	Encoder–Decoder + MCA + DAT	Combines multi-dimensional convolutional attention and a dual transformer.	Fragility in very dynamic scenes.
Hybrid Attention (Liu et al.) [115]	2024	Self-supervised	Transformer + Hybrid Attention	Refines multi-resolution maps combining global attention and local convolutions.	More complex and heavier training.
PAR-mono [116]	2024	Self-supervised (video)	Adaptive Encoder + Dynamic Attention	Improves temporal consistency and accuracy in video.	Dependent on stable sequences.

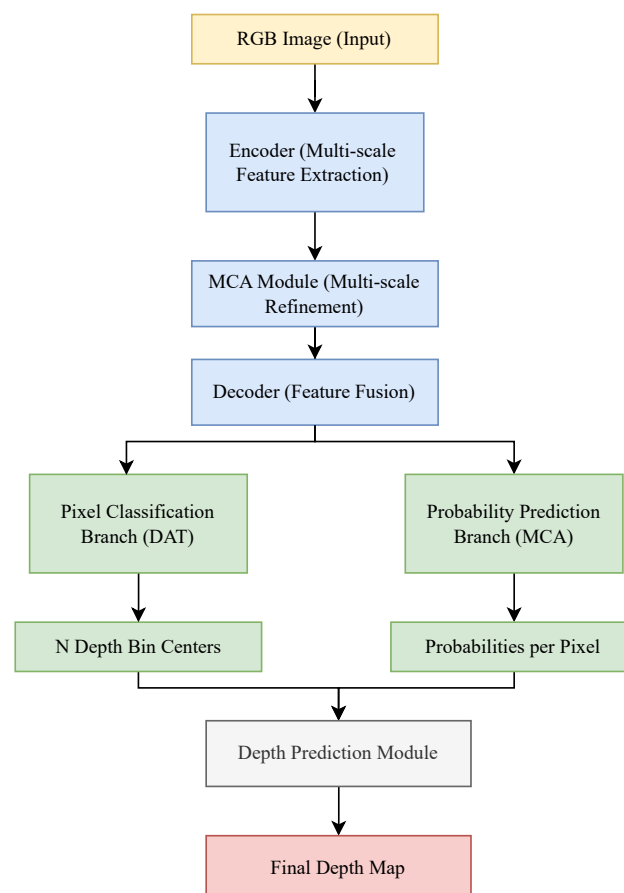


Figure 12. Diagram of the CATNet architecture based on [114].

3.3. End-to-End SLAM: From Deep Visual Odometry to Learned Mapping

The evolution of Visual and Visual-Inertial SLAM approaches driven by Deep Learning has increasingly led to integrable and differentiable systems. In modular approaches, neural networks replace specific components of a given pipeline (such as feature detection or depth estimation), whereas end-to-end approaches aim to learn directly the correspondence between image sequences and the underlying trajectories/mappings, uniquely enabling motion estimation, 3D reconstruction, sensor fusion, and geometric optimization. Figure 13 provides a historical conceptualization of the evolution of VSLAM and VI-SLAM with Deep Learning.

This transition represents a profound conceptual shift: the focus moves from replacing individual blocks to learning the set of poses, depth, and the map within a single model. In this way, the system learns implicit geometric representations of the scene, simultaneously optimizing both the network parameters and the photometric consistency across views. In general, end-to-end approaches can be seen as a natural extension of learned Visual Odometry (VO) networks, which were the initial steps toward integrating visual perception and motion estimation within the same model.

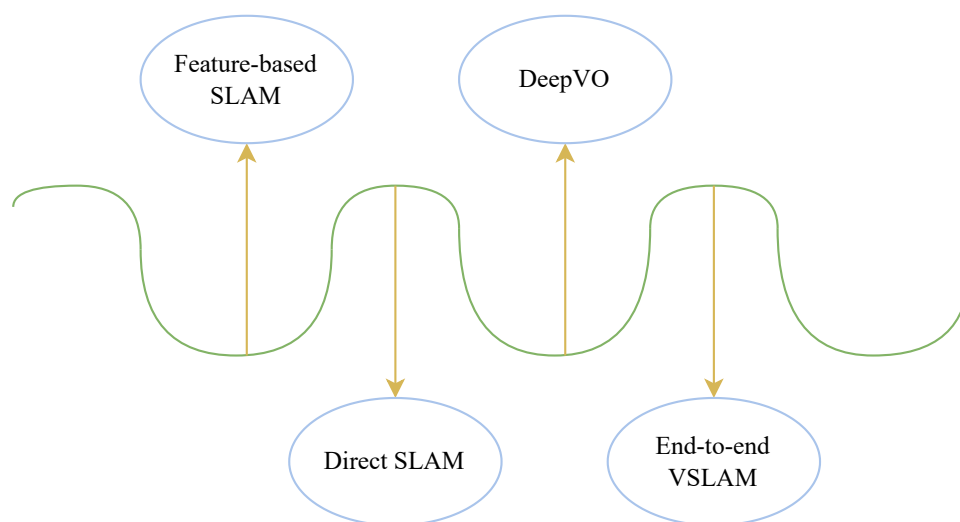


Figure 13. Temporal Contextualization of the Evolution of SLAM to Deep-Learning-based SLAM.

The first efforts to integrate neural networks into the VSLAM pipeline focused on replacing classical Visual Odometry (VO) methods with Deep Learning models. These methods, known as Deep Visual Odometry (DeepVO), aimed to map image sequences directly to camera trajectories using CNNs and recurrent neural networks (RNNs/LSTMs) [117]. This type of approach, as illustrated in Figure 14, marked the beginning of the transition from purely geometric pipelines to the first differentiable systems.

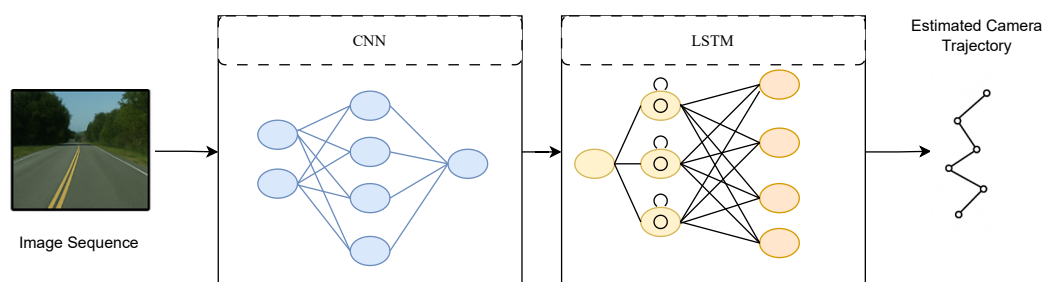


Figure 14. The DeepVO framework, consisting of an image sequence that subsequently passes through the CNN layer, then the LSTM layer, and finally the estimated trajectory is obtained. Based on [117].

One of the first supervised models was DeepVO [117], where a CNN extracts spatial features and an LSTM models the temporal dependency between consecutive frames. Following this, VINet [118] built on the DeepVO concept but extended it by integrating IMU data, using a multimodal CNN+LSTM architecture to improve robustness in low-texture scenarios. Subsequently, approaches exploring self-supervised learning emerged. UnDeepVO [119], a variant of DeepVO, introduced unsupervised training through photometric consistency, while GANVO [120] combined Generative Adversarial Networks (GANs) to reconstruct synthetic views and improve training stability.

Hybrid methods also began to emerge, attempting to combine neural networks with geometric components. For instance, CNN-SVO [28] integrates CNNs for depth estimation with the classical SVO algorithm [52] for pose estimation, representing a semi-learned system. D3VO [121] combined CNNs with traditional photometric optimization to improve temporal consistency and motion smoothness. In the visual-inertial context, DeepVIO [122] and VIOLearner [123] performed learned fusion of camera and IMU data. The former adopted a supervised CNN+LSTM stereo architecture to capture spatio-temporal relations, while the latter introduced a self-supervised mechanism capable of learning implicit sensor calibration.

More recently, approaches have evolved into fully end-to-end pipelines, as illustrated in Figure 15, where feature extraction, pose estimation, and map reconstruction are modeled as a single differentiable process.

In this direction, Li R. et al. [124] developed DeepSLAM, representing an important milestone in the transition from hybrid systems to truly end-to-end solutions. This method proposes a unified neural network capable of integrating feature extraction, matching, and pose optimization within a single differentiable workflow. The architecture combines CNNs for visual perception and RNNs for temporal reasoning, enabling continuous and coherent camera trajectory estimation over long sequences. In 2021, LIFT-SLAM [125] deepened this integration by incorporating the concept of Local Image Features with Transformers, combining feature extraction and keypoint matching in a unified attentional architecture. The model uses multi-head transformers to capture long-range dependencies between frames, improving robustness against blur, repetitive textures, and viewpoint variations. These methods were later followed by other architectures, such as DROID-SLAM [16], one of the first to introduce a recurrent attention-based architecture that unifies dense matching, flow estimation, and iterative optimization, becoming one of the first end-to-end systems to outperform classical methods like ORB-SLAM3 in dynamic scenarios. Similarly, DVI-SLAM [126], proposed by Peng X. et al., integrates a dual visual-inertial fusion architecture with two parallel networks: one for image processing and one for the IMU. The networks interact through cross-attention mechanisms, achieving superior robustness under vibrations and variable lighting. D-VINS [127] introduced a Dynamic Adaptive Visual-Inertial SLAM paradigm, integrating a dynamic inertial prior and semantic constraints to handle highly mobile environments. This system reflects an emerging trend: the fusion of semantics and sensor dynamics, where networks learn to distinguish between moving and static objects, adaptively adjusting pose estimation. In 2024, a real-time visual-inertial framework called SuperVINS [17] was proposed, capable of operating under adverse imaging conditions. Its architecture combines SuperPoint-based feature extraction with a differentiable optimization module, allowing online adjustments to sensor uncertainties. Figure 15 shows the architecture of this pipeline in detail. More recently, LET-SE2-VINS [128] introduced a hybrid structure based on learned optical flow and SE(2) geometry, designed to improve the robustness of visual-inertial fusion under vibrations and partial occlusions. LVID-SLAM [129] presented a lightweight approach to Visual-

Inertial SLAM in dynamic environments, combining semantic segmentation for moving object detection with an architecture optimized for edge-device deployment.

Thus, Table 11 presents a comparison of the various algorithms that combine neural networks with VSLAM and VI-SLAM.

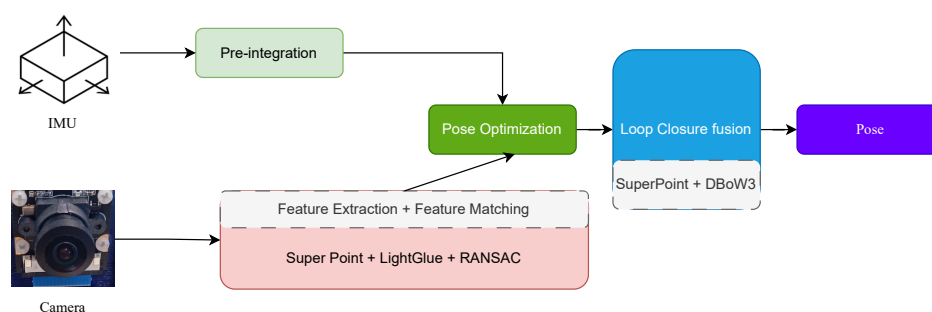


Figure 15. SuperVins end-to-end pipeline architecture. Based on [17].

Table 11. Main end-to-end algorithms and frameworks that integrate neural networks in Visual and Visual-Inertial SLAM.

Method	Year	Sensor	Neural Architecture	Supervision	Main Contributions
DeepVO [117]	2017	Mono	CNN + LSTM	Sup.	First Deep-Learning-based VO.
VINet [118]	2017	Mono+IMU	CNN + LSTM	Sup.	First learned visual-inertial fusion with multimodal integration.
UnDeepVO [119]	2018	Mono	VGG Enc-Dec	Unsup.	Self-supervised training via photometric consistency and scale recovery.
GANVO [120]	2019	Mono	CNN + GAN	Unsup.	Synthetic reconstruction via GAN with increased training stability.
CNN-SVO [28]	2019	Mono	CNN + Geom.	Hybrid	Combines learned depth with classic SVO.
D3VO [121]	2020	Mono	CNN + Diff. BA	Hybrid	Learning of photometric uncertainty and adaptive optimization.
DeepVIO [122]	2019	Stereo+IMU	CNN + LSTM	Sup.	Supervised visual-inertial fusion with dual architecture.
VIO Learner [123]	2020	Mono+IMU	CNN	Unsup.	Learns implicit calibration between sensors.
DeepSLAM [124]	2021	Mono	CNN + RNN	Hybrid	Unified end-to-end pipeline that integrates feature extraction and optimization.
LIFT-SLAM [125]	2021	Mono	Transformer	Sup.	Combines spatial and temporal attention; differentiable geometric optimization.
DROID-SLAM [16]	2021	Mono/Stereo	Recurrent Transformer	Self-sup.	Recurrent attention that replaces classic VO/SLAM modules.
DVI-SLAM [126]	2023	Mono+IMU	Dual CNN + Cross-Att.	Sup.	Cross-attention based visual-inertial fusion.
SuperVINS [17]	2023	Mono+IMU	SuperPoint + Diff. Opt.	Hybrid	Real-time adaptive SLAM under low illumination.
D-VINS [127]	2024	Mono+IMU+Sem.	CNN + Attention + Sem.	Hybrid	Inertial priors and semantic constraints for dynamic environments.
LET-SE2-VINS [128]	2025	Mono+IMU	Learned Flow + SE(2)	Hybrid	Robust fusion under vibrations and partial occlusions.
LVID-SLAM [129]	2025	Mono+IMU+Sem.	Lightweight Semantic CNN	Hybrid	Lightweight and semantic SLAM for edge devices in dynamic environments.

3.4. Semantic and Contextual SLAM

The integration of semantics and context into VSLAM and VI-SLAM systems represents a significant evolution driven by Deep Learning. Semantic SLAM refers to a system that not only obtains geometric information about an unknown environment and the movement of a UGV but also detects and identifies objects within that environment [11]. This transition from purely geometric to semantic and contextual understanding is crucial in complex and unstructured environments, such as agricultural and forested areas, where vegetation, variable lighting, and dynamic objects challenge the assumptions of traditional methods.

Traditional VSLAM describes the environment using point clouds and other geometric representations, which alone constitute only a set of points devoid of semantic meaning.

For robots to perceive and interact with the world in a manner closer to human understanding, it is essential that they can abstract and assign meaning to these points, interpreting the content and context they represent [11]. In this way, Semantic and Contextual SLAM aims to assign meaning to visual observations by incorporating neural networks for semantic segmentation, object detection, and contextual reasoning. Consequently, the map becomes a rich representation of the environment, where each element is identified and classified, enabling more intelligent interaction between a UGV and the world. Figure 16 shows a diagram comparing Classical VSLAM with Semantic SLAM.

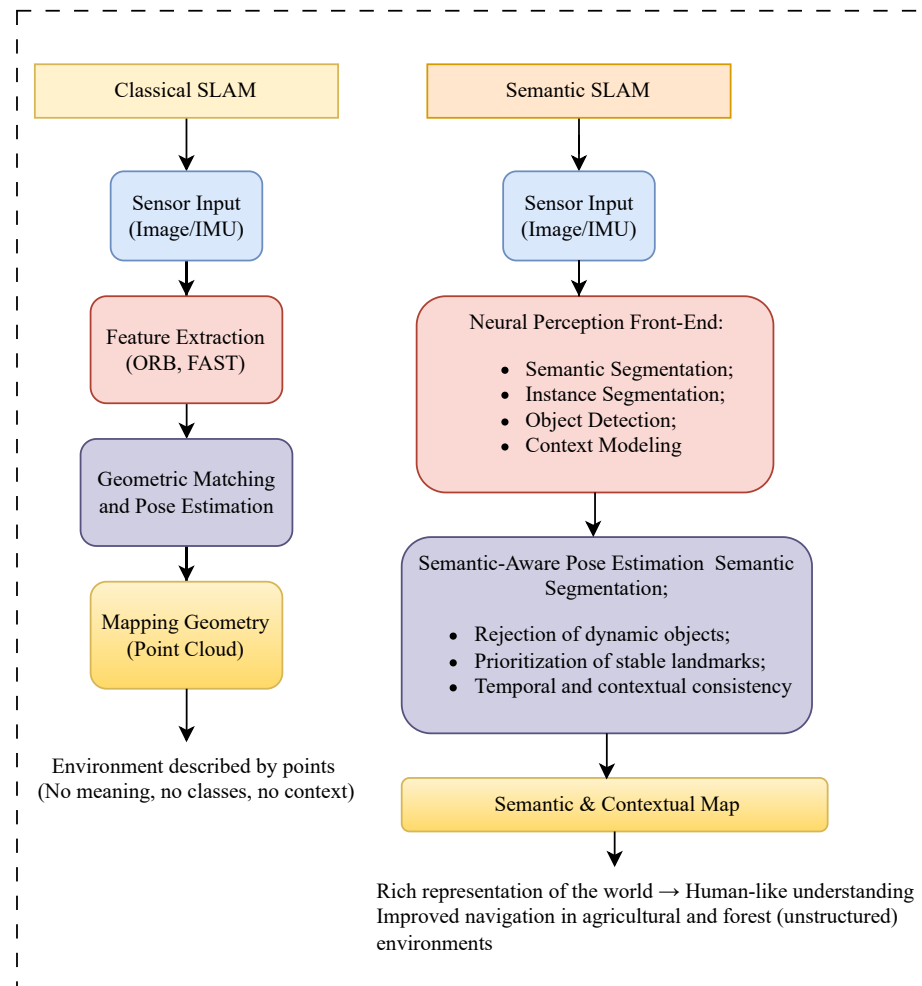


Figure 16. Classical VSLAM vs. Semantic VI-SLAM.

3.4.1.1. Dynamic and Contextual Object-Aware SLAM

One of the main differences between traditional VSLAM and semantic VSLAM lies in the integration of object detection (and characterization) mechanisms, allowing the assignment of attributes and semantic data to objects present in a given environment.

The classification and interpretation of visual elements in a scene thus represents one of the first steps in semantic VSLAM. For example, in dynamic environments such as agricultural or forested areas, the integration of object detection and segmentation algorithms provides a significant advantage. In this context, object detection algorithms based on Deep Learning typically follow the following structure [130,131]:

- Backbone, responsible for extracting the main features from the image, serving as the foundation of the network.
- Neck, functions to fuse and refine the extracted features, ensuring a better representation of the information before the final detection stage.

- Head, the module responsible for predicting object classes and estimating their positions (coordinates) in the image.

Several algorithms have emerged from this structure, with notable examples being the R-CNN series [132], SSD [133], and YOLO [134], which provided significant improvements in performance and speed compared to previous methods.

Recently, with the advent of Transformers, the computer vision paradigm gained new dynamism due to the adaptive parameter-learning capabilities of neural networks. The introduction of the Vision Transformer (ViT) [135] enabled the adaptation of Transformer architecture for vision tasks by dividing an image into patches, which are then treated as a sequence of tokens. However, due to the high computational cost of ViT, more efficient variants emerged, notably DeiT [136] and Pyramid Vision Transformer (PVT) [137], which incorporated hierarchical multi-scale structures similar to CNNs. Among these evolutions, the Swin Transformer [138] stands out, using local attention based on shifted windows, offering an ideal balance between accuracy and computational efficiency. This architecture has been widely adopted in robotic perception tasks and integrated into pipelines such as SwinDepth [113] for depth estimation and CAT-SLAM. Segmentation-oriented models like SegFormer [139] and universal architectures like the Segment Anything Model (SAM) [140] have further expanded visual perception capabilities, enabling adaptive segmentation of any object, even without specific supervised training. These models are now utilized in Dynamic Object-Aware SLAM systems, such as D-VINS [127] and LVID-SLAM [129], where they assist in detecting and filtering moving objects, improving pose estimation robustness.

In summary, despite advances in neural networks and Transformer architectures, the efficient real-time extraction and integration of semantic information remains a critical challenge. At the same time, it represents a decisive step toward the development of more intelligent, robust, and adaptive VSLAM systems. Table 12 presents a comparison of the contributions of each algorithm discussed in this subsection.

3.4.2. Semantic Mapping

One of the main convergence themes between VSLAM/VI-SLAM and Deep Learning is the construction of semantic maps. For a UGV to robustly interpret an environment in a human-like manner and subsequently perform complex tasks in different contexts, it needs to go beyond geometric representation. Unlike traditional maps based on point clouds or occupancy grids, Semantic Maps include labels and contextual relationships between objects, surfaces, and regions of the scene, enabling high-level perception essential for autonomous decision-making.

With the introduction of CNNs and Transformers, architectures have been established that allow the combination of geometric, visual, and semantic information in an end-to-end manner. Models such as Panoptic-DeepLab [141] and Mask2Former [142] enable real-time panoptic segmentation, simultaneously distinguishing individual objects (instances) and continuous regions (global semantics).

More recently, emerging methods based on Neural Implicit Maps (NIMs), such as Semantic-NeRF [143] and GSNeRF [144], leverage Neural Radiance Fields (NeRFs) to model the geometry and semantic content of a scene in a 3D latent space. These approaches allow dense fusion between visual reconstruction and semantic segmentation, producing detailed and consistent 3D maps even in partially observed environments. Hybrid architectures have also appeared, such as SplatAM [145] and S-NeRF-SLAM [146] in 2023, which combine NeRFs with differentiable geometric optimization, enabling the creation of continuous 3D maps that are also semantically rich directly from video sequences. These systems overcome the limitations of classical voxel-based methods, offering scalability and generalization capability in complex environments, such as forests or agricultural fields.

Table 12. Main Transformer architectures applied to object detection and segmentation and their integration into VSLAM/VI-SLAM pipelines.

Model/Architecture	Year	Main Characteristics	Integration/Application in SLAM
Vision Transformer (ViT) [135]	2020	Global attention over <i>patches</i> and models long-range dependencies	Base for ViT-SLAM and ViT-Depth models, used in global scene perception
DeiT [136]	2021	Efficient ViT variant with <i>knowledge distillation</i> and requires less training data	Experimental applications in <i>feature extraction</i> and hybrid CNN+Transformer networks
PVT (Pyramid Vision Transformer) [137]	2021	Multi-scale hierarchical structure that preserves local and global information	Used as a <i>backbone</i> in segmentation and semantic mapping pipelines
Swin Transformer [138]	2021	Hierarchical attention based on <i>shifted windows</i> and efficient in real-time	Integrated into SwinDepth and CAT-SLAM, and robust in natural environments and variable illumination
SegFormer [139]	2021	Transformer Encoder + lightweight decoder with high accuracy and low computational cost	Used for semantic segmentation in <i>Dynamic Object-Aware SLAM</i> systems
Segment Anything Model (SAM) [140]	2023	Universal attention-based segmentation and detects objects without supervised training	Integrated into <i>Dynamic Object-Aware SLAM</i> pipelines for adaptive filtering of moving objects

Note: D-VINS [127] and LVID-SLAM [129] systems explore these architectures for dynamic object detection and filtering, but were not included in the table as they focus on visual–inertial sensor fusion.

4. Applications and Challenges in Agriculture and Forestry Robotics

The integration of technological solutions in the field of autonomous systems has revolutionized several sectors of society. Their use has shown a significant impact, marking the beginning of a new era in robotics, helping humans perform dangerous activities, and protecting them from environments considered hazardous and difficult to operate in. Among these environments, unstructured natural settings stand out, such as those related to the agricultural and forestry sectors.

4.1. Challenges

Unstructured natural environments encompass a wide spectrum of application contexts, ranging from semi-structured agricultural fields to highly unstructured forested areas. While both domains share challenges such as vegetation-induced occlusions and variable lighting, they differ substantially in terms of motion constraints, environmental dynamics, and sensing reliability. This section therefore distinguishes between agricultural and forestry scenarios, highlighting how different SLAM architectures and Deep Learning strategies align with the specific requirements of each domain.

Autonomous navigation of mobile robots in unstructured natural environments, such as forested and/or agricultural terrains, faces significant challenges compared to urban or industrial settings. These environments are not modified for UGV operations, and their

constant dynamics create uncertainties, requiring robots to possess adaptive capabilities in their tasks [147].

However, progress in this field is accompanied by a set of complex challenges that continue to limit its full development. Among the main obstacles are the inherent difficulties of navigating unstructured terrains, which demand systems capable of handling irregularities, slopes, and dynamic obstacles. A critical factor is the reliability of localization and consistent mapping, which can sometimes be degraded due to traction systems losing stability on uneven ground and limitations of GNSS systems in densely vegetated areas that compromise precise signal coverage [148].

In unstructured environments, there are some factors that directly affect the performance of VSLAM and VI-SLAM systems. Among these, the most challenging are the presence of dynamic obstacles, unpredictable/adverse environmental variations [5,6], and occlusion caused by environmental elements outside the sensor's field of view [149]. Additionally, perception complexity in irregular terrains generates problems such as visual aliasing, where similar patterns make feature matching between visual sensors difficult [150]. Inertial sensors, on the other hand, suffer from drift over time, exacerbated by vibrations and uneven terrain [151]. Consequently, sensor fusion between cameras, IMUs, and sometimes LiDAR or GNSS becomes essential, although its reliability remains limited in densely vegetated forested or agricultural areas [152].

Overcoming these challenges has been driven by the introduction of Deep-Learning-based techniques, which enable greater resilience to environmental variations and more robust semantic feature extraction. This paradigm has opened new perspectives in both robust navigation and contextual interpretation, creating a convergence point between classical SLAM and adaptive semantic SLAM, particularly relevant in agricultural and forestry domains [153].

4.2. *Advances in Agriculture*

With the continuous growth of the global population, there has been increasing pressure for agricultural systems that maximize certain tasks more efficiently and sustainably [154]. Modern techniques, such as precision agriculture, employ advanced technologies such as GNSS systems, UAVs, and AI methods to optimize resource usage and maximize productivity [155].

Agriculture 4.0 and 5.0 strategies represent significant evolutions in the sector, combining efficiency, sustainability, and technological innovation. Agriculture 4.0 emphasizes the automation of operational tasks, whereas Agriculture 5.0 integrates advanced robotics, autonomous UAVs, and intelligent sensors, promoting productivity, cost reduction, and sustainable practices [155,156].

In this context, agricultural robotics plays a central role by automating planting, fertilization, harvesting, and crop mapping operations with high precision. It is noteworthy that agricultural applications range from open field scenarios to semi-structured and confined environments, such as greenhouses and cultivation corridors. In these environments, domain-specific perception and control strategies can be effectively exploited. For example, Tsiakas et al. [157] proposed an autonomous navigation framework for holonomic mobile robots operating in confined agricultural environments, combining customized perception modules with robust navigation strategies validated on real robotic platforms. Another example is Chen et al. [158] presented a methodology for large-scale, unstructured orchard 3D mapping, combining a stereo vision system mounted on a robotic arm with SLAM techniques. The system captures images from multiple angles, generating accurate 3D maps of irregular orchards, even with dense vegetation, providing valuable data for agricultural automation and planning. In [159], the authors proposed a low-cost, energy-efficient local

motion planner for autonomous navigation of vehicles in vineyard rows, using only an RGB-D camera and simple hardware. The system employs a two-layer algorithm: the first layer leverages the camera's depth (disparity) map to generate direct proportional control of the vehicle, while the second layer functions as a backup, using a Deep Learning model (CNN) trained via transfer learning to generate motion primitives in case of primary layer failure. Another widely recognized system, developed in 2023, is AGRI-SLAM, a real-time stereo visual SLAM system designed for challenging agricultural environments. It utilizes image enhancement and ORB feature points with LSD line extraction to operate under low-light and fog conditions. Tested on standard datasets and real orchards, it outperformed methods such as ORB-SLAM2, VINS-SLAM, and PL-SLAM, providing accurate localization and mapping even where GNSS reliability is limited [153]. Figure 17 illustrates the platform used in AGRI-SLAM, the environment in which the UGV operated, and the obtained results.

In [160], the authors developed a UGV integrating tightly coupled fusion of GNSS, stereo vision, and IMU measurements within the ORB-SLAM3 pipeline for autonomous navigation and mapping in agricultural fields. Experiments conducted in soybean fields (using the Rosario Dataset and proprietary data) reported pose error reductions of approximately 10% to 30% compared to visual-inertial approaches and loosely coupled GNSS–stereo–inertial fusion methods. The study addressed typical challenges of such scenarios, including repetitive visual appearance, difficult terrain, vibrations, and GNSS signal loss or degradation. In [161], the authors presented a monocular visual-inertial odometry (VIO) system specifically adapted for challenging agricultural environments characterized by irregular terrain, dense vegetation, and variable lighting. Based on VINS-Mono, the method incorporates improvements to the vision front-end, keyframe selection to avoid tracking loss, pose optimization with filtering of low-depth points for higher accuracy, and a failure recovery module that resumes trajectory after temporary tracking loss. Tests in agricultural fields using the Rosario dataset showed an absolute trajectory error reduction of up to 69% compared to the original VINS-Mono. Recently, Tran Q. et al. [162] proposed a monocular visual-inertial odometry system dedicated to agricultural vehicles, integrating monocular camera and IMU measurements with the correction of 3D feature points augmented by the IMU. The approach uses the IMU to assist triangulation or correction of 3D feature points to reduce the effects of irregular terrain, vibrations, and visual instabilities typical in agricultural environments. In particular, upon detecting regions of instability or increasing error, the system applies automatic adjustment of 3D feature points based on IMU acceleration/rotation data, enhancing scale and trajectory consistency. In [163], the authors proposed a mapping algorithm called motion-coupled mapping, aimed at characterizing the canopy structure of hybrid rice in agricultural fields, focusing on autonomous ground agricultural vehicles (Agri-UGVs) operating in relatively complex and unstructured paddy environments. The algorithm integrates real-time RGB-D sensor data (image + depth) with vehicle kinematic and inertial measurements (IMU/odometry).

In summary, recent advances in sensors, computer vision, SLAM, and robotics are transforming agriculture, enabling precise automation, detailed 3D mapping, and autonomous navigation in complex scenarios. The integration of visual, inertial, and GNSS systems, combined with artificial intelligence algorithms, demonstrates significant potential to increase productivity, reduce costs, and enhance sustainability across different crops and terrain types, consolidating the technological foundation for Agriculture 4.0 and 5.0. Table 13 presents a comparison of the various methods discussed.



Figure 17. AGRI-SLAM System. Data extracted from [153], licensed under CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>): (a): Robotic platform (UGV). (b): Demonstration of the UGV's operation scenario.

Table 13. Comparison of robotics and SLAM methods in agricultural environments.

Method	Sensors	Algorithm	Main Characteristics	Advantages/Results
Tsiakas et al. [157]	RGB-D + LiDAR	Semantic navigation framework	Autonomous navigation in confined and semi-structured agricultural environments using domain-specific perception and control strategies	Robust navigation validated on real platforms. Exploits environmental structure for improved localization and safety
Chen et al. [158]	Stereo camera (robotic arm)	3D Visual SLAM	3D mapping of large, unstructured orchards.	Accurate maps of irregular orchards, useful for automation and agricultural planning
Aghi et al. [159]	RGB-D camera	Local Planner + CNN backup	Autonomous navigation in vineyard rows with a CNN architecture that acts as a backup in case of sensor failure	Low-cost, energy-efficient system, and robust to perception failures
AGRI-SLAM [153]	Stereo camera	Real-time Stereo Visual SLAM	Operates in low light and fog with image enhancement and ORB + LSD extraction	Outperforms ORB-SLAM2 and VINS-SLAM with robust localization even with limited GNSS
Cremona et al. [160]	GNSS + Stereo + IMU	ORB-SLAM3 with tight fusion	Navigation in soybean fields with an error reduction of 10–30%. Handles difficult terrain and GNSS signal loss	Higher accuracy than loosely coupled fusions and robust in complex environments agricultural scenarios
Song et al. [161]	Monocular VIO + IMU	Enhanced VINS-Mono	Keyframe selection, low-depth filtering, failure recovery module; suitable for dense vegetation	Trajectory error reduction up to 69% and stable localization in uneven terrain
Tran et al. [162]	Monocular VIO + IMU	3D Point Correction via IMU	Automatic adjustment of 3D points using inertial data and reduces visual drift	More consistent and robust trajectories, suitable for agricultural UGVs
Feng et al. [163]	RGB-D + IMU/odometry	Motion-Coupled Mapping	Hybrid rice field characterization.	Accurate canopy mapping; support for Agri-UGVs in complex rice fields

4.3. Advances in Forestry

The integration of systems capable of assisting workers in the forestry industry has become a critical topic in this field due to the risks associated with wildfires, whose frequency

and severity have increased as a result of climate change, environmental preservation, and monitoring [164].

Forested environments represent one of the most challenging scenarios in unstructured natural environments for UGV operations. Unlike organized urban and/or agricultural areas, forests exhibit higher vegetation density, variable lighting, occlusions, and the presence of dynamic obstacles [165]. These characteristics lead to significant degradation in SLAM methods (whether LiDAR-based, VSLAM, or VI-SLAM), prompting the scientific community to increasingly develop new approaches that are more robust, multimodal, and adaptive for autonomous navigation in forest contexts.

In recent years, various projects and robotic platforms have emerged specifically for forest navigation, testing different sensor modalities and sensory fusion mechanisms. A notable example is Tree-SLAM [165], which exploits tree trunk detection as stable landmarks to provide consistent mapping even in dense vegetation conditions. By transforming trunks into persistent geometric entities, this method mitigates the loss of visual features caused by foliage, shadows, or repetitive patterns typical of dense forests. Another relevant work by Pierzchała et al. [166] demonstrated the capability of a UGV equipped with 3D LiDAR and Graph-SLAM to reconstruct forest environments with high accuracy, proving particularly useful for forest inventory. The use of high-resolution LiDAR sensors allowed the system to overcome challenges related to poor lighting and texture, providing detailed 3D maps even under adverse conditions. Faitli et al. [167] integrated a Mobile Laser Scanning (MLS) system combining LiDAR and IMU into a forest harvester for positioning under canopy coverage and automatic estimation of tree attributes (DBH, trunk curve). This study illustrates how multimodal SLAM (laser + inertial) can support real forestry operations, highlighting the need for robust sensor fusion in dense natural environments. A more comprehensive system is LVI-ObjSemantic [168], which combines LiDAR, vision, and IMU in a semantic SLAM pipeline aimed at identifying and mapping individual trees. The method employs segmentation networks to extract trunks and integrates visual and geometric information into an efficient hierarchical structure (Cluster-Block-Single/Global), reducing false positives and stabilizing frame-to-frame correspondence. More recently, Shi K. et al. [169] proposed a method called YSLK-SLAM specifically designed for “urban forests.” The system extends ORB-SLAM2 with a semantic segmentation layer based on YOLOv8-seg, enabling the identification and filtering of dynamic objects (such as people or animals), preventing them from introducing pose estimation errors. To reinforce the semantic mask, the authors further integrate a failed-detection module and a depth-difference-based criterion.

In summary, recent advances in multimodal sensors, computer vision, SLAM, and Deep Learning are significantly transforming forestry robotics. Dense and unstructured environments require robust solutions combining LiDAR, cameras, and IMUs. Modern systems integrate geometric, semantic, and dynamic object filtering approaches, allowing for tree trunk identification, individual tree mapping, and the execution of tasks such as forest inventory, trail planning, logging operation monitoring, and environmental surveillance. The incorporation of Deep Learning techniques, such as semantic segmentation, trunk detection, and transformer-based models, is significantly improving perception quality under adverse conditions, thus revolutionizing a new generation of forestry robotics. Table 14 provides a comparison of the methods presented in this subsection.

Table 14. Main SLAM methods applied to forest environments.

Method	Year	Sensors	Key Idea	Contribution
Tree-SLAM [165]	2015	RGB/Depth Camera	Detection of tree trunks as stable cylindrical landmarks	Robust mapping in dense vegetation and reduces visual losses caused by foliage and shadows
Pierzchała et al. [166]	2018	3D LiDAR	Graph-SLAM based on dense clouds	Accurate forest reconstruction and useful for forest inventory
Faitli et al. [167]	2020	LiDAR + IMU (MLS)	Harvester localization and automatic tree measurement	Demonstrates multimodal SLAM for real forest operations
LVI-ObjSemantic [168]	2023	LiDAR + Camera + IMU	Semantic SLAM based on trunk segmentation	Reduces false positives with mapping oriented towards individual trees
YSLK-SLAM [169]	2025	RGB + IMU	ORB-SLAM2 + YOLOv8-seg + dynamic masks	Robust SLAM in “urban forests”; filters people/animals and reinforces pose stability

5. Conclusions and Future Perspectives

In recent years, SLAM has emerged as one of the primary challenges in robotics, particularly in unstructured natural environments such as forests and irregular agricultural terrains. In these scenarios, autonomous navigation is hindered by factors such as dense vegetation, uneven terrain, variable lighting, and unpredictable obstacles, demanding robust and precise perception systems. Vision-based technologies, such as VSLAM and VI-SLAM, have been extensively developed to address these needs and have been successfully applied in UGVs.

This manuscript presents a critical survey of both classical and modern Visual SLAM (VSLAM) and Visual-Inertial SLAM (VI-SLAM) approaches applied to UGVs operating in unstructured natural environments. The theoretical foundations of feature-based, direct, and hybrid methods were reviewed, framing the architectures of tight-coupled versus loosely coupled visual-inertial fusion.

With the advent of AI, higher demands have emerged regarding the autonomy of mobile robots, particularly concerning advanced environmental perception. To meet these requirements, Deep Learning techniques have been developed, enabling systems not only to reconstruct geometric features, as in traditional VSLAM, but also to incorporate higher-level environmental information. Although deep reinforcement learning has demonstrated promising results for autonomous navigation, such approaches typically rely on accurate state estimation and mapping as underlying components. In many practical systems, learned navigation policies are trained and executed on top of SLAM-based localization or assume the availability of a consistent spatial representation. Visual and Visual-Inertial SLAM therefore remain fundamental for providing explicit geometric consistency, reliable localization, and map reuse, which are essential for long-term autonomy and safety-critical operation. This is particularly relevant in large-scale outdoor environments, where purely end-to-end navigation policies often struggle to generalize beyond training conditions due to environmental variability and limited interpretability. Recent advances in Deep Learning focus on four main areas: learned feature detection and matching, depth estimation (including self-supervised approaches), end-to-end pipelines, and semantic/contextual mapping. Current datasets and benchmarks for agro-forestry scenarios were evaluated, and literature solutions addressing the dynamics and complexity of these environments

were discussed. Overall, recent literature demonstrates consistent progress but also highlights clear weaknesses in generalization, robustness under vegetation cover, and the lack of specific multimodal datasets for in-field validation. However, with advances in Deep Learning and computational performance, it has become possible to detect and classify objects directly from pre-trained images in real-time, increasing both generalization capability and practical relevance for semantic VSLAM systems.

The main contribution of this survey is to provide a consolidated view of classical and emerging VSLAM and VI-SLAM techniques, focusing on UGVs in agro-forestry environments. It highlights critical comparisons between feature-based versus direct methods, tight-coupled versus loosely coupled architectures, and learning-based modules. The evolution of VSLAM and VI-SLAM has significantly improved robustness, accuracy, and high-level perception, enhancing the ability of robots to interact autonomously in unstructured natural environments. Unstructured natural environments encompass a wide range of application contexts, including agricultural fields and forested areas. Each setting introduces distinct sensing constraints, motion assumptions, and learning requirements. For instance, agricultural and forest environments often benefit from domain-specific perception strategies and semantic representations, while construction sites and semi-structured outdoor environments demand robustness to dynamic obstacles and incomplete prior maps. This diversity highlights that no single VSLAM solution generalizes across all unstructured environments, reinforcing the need for application-driven system design.

Overall, the reviewed literature suggests that the effectiveness of VSLAM and VI-SLAM systems is highly dependent on the target environment and task. Approaches that integrate learning-based perception with explicit geometric representations appear particularly promising for real-world deployments in unstructured natural environments. Nevertheless, challenges related to generalization across domains, dataset availability, and long-term robustness under vegetation, lighting changes, and dynamic conditions remain open.

The analysis identified key gaps guiding future research, such as the need for multimodal datasets with reliable ground-truth under vegetation cover, adaptive sensor fusion mechanisms, and learning strategies that generalize across different domains and environmental conditions. Based on this review, several research priorities for strengthening VSLAM/VI-SLAM in unstructured natural environments can be outlined. First, sensor interaction emerges as a central theme, requiring the development of mechanisms that dynamically adjust the contribution of each sensor (monocular or stereo camera, IMU, GNSS/RTK) according to local reliability, reducing reliance on fixed linear models and enhancing resilience to temporary failures. In parallel, the development of self-supervised and domain-adaptive methods for monocular depth estimation will allow models to be trained directly in the field without LiDAR or RTK ground-truth, ensuring transfer across different vegetation types, seasons, and lighting conditions.

Another promising direction is semantic integration and dynamic object filtering, enabling VSLAM systems to identify and exclude moving or irrelevant elements, such as people, animals, or moving foliage, thereby improving backend consistency and mapping robustness in complex natural environments. Combining geometric mapping with detailed semantic maps is particularly relevant for agro-forestry applications. Additionally, hybrid architectures that integrate learned modules (frontends) with traditional graph-based optimization can provide an effective balance between performance, interpretability, and computational efficiency, which is essential for deploying these methods on edge devices.

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References

1. Dhanda, M.; Rogers, B.A.; Hall, S.; Dekoninck, E.; Dhokia, V. Reviewing human-robot collaboration in manufacturing: Opportunities and challenges in the context of Industry 5.0. *Robot. Comput.-Integr. Manuf.* **2025**, *93*, 102937. [\[CrossRef\]](#)
2. Chen, Z.; Du, Y.; Xiao, J.; Sun, T.; Wang, Z.; Eynard, B.; Zhang, Y. Semantic map construction approach for human-robot collaborative manufacturing. *Robot. Comput.-Integr. Manuf.* **2025**, *91*, 102845. [\[CrossRef\]](#)
3. Lopes, D.; Pereira, T.; Gonçalves, A.; Cunha, F.; Lopes, F.; Antunes, J.; Santos, V.; Coutinho, F.; Barreiros, J.; Durães, J.; et al. Integrated Fleet Management of Mobile Robots for Enhancing Industrial Efficiency: A Case Study on Interoperability in Multi-Brand Environments Within the Automotive Sector. *Appl. Sci.* **2025**, *15*, 7235. [\[CrossRef\]](#)
4. Beyçimen, S.; Ignatyev, D.; Zolotas, A. A Comprehensive Survey of Unmanned Ground Vehicle Terrain Traversability for Unstructured Environments and Sensor Technology Insights. *Eng. Sci. Technol. Int. J.* **2023**, *47*, 101457. [\[CrossRef\]](#)
5. Naderi, H.; Shojaei, A.; Huang, L. Foundation Models for Autonomous Robots in Unstructured Environments. *arXiv* **2024**. [\[CrossRef\]](#)
6. Bavle, H.; Sanchez-Lopez, J.L.; Cimorelli, C.; Tourani, A.; Voos, H. From SLAM to Situational Awareness: Challenges and Survey. *Sensors* **2023**, *23*, 4849. [\[CrossRef\]](#)
7. Durrant-Whyte, H.; Bailey, T. Simultaneous Localization and Mapping: Part I. *IEEE Robot. Autom. Mag.* **2006**, *13*, 99–110. [\[CrossRef\]](#)
8. Ding, H.; Zhang, B.; Zhou, J.; Yan, Y.; Tian, G.; Gu, B. Recent Developments and Applications of Simultaneous Localization and Mapping in Agriculture. *J. Field Robot.* **2022**, *39*, 956–983. [\[CrossRef\]](#)
9. Zhou, T.; Fang, R.; Wang, F.; Wang, G.; Wu, L.; Su, K.; Xu, L.; Pu, H.; Luo, J. Visual and LiDAR SLAM Performance in Campus Scenes. In Proceedings of the 2024 3rd International Conference on Service Robotics (ICoSR 2024), Hangzhou, China, 26–28 July 2024; pp. 83–88. [\[CrossRef\]](#)
10. Huang, L. Review on LiDAR-based SLAM Techniques. In Proceedings of the 2021 International Conference on Signal Processing and Machine Learning (CONF-SPML), Stanford, CA, USA, 14 November 2021; pp. 163–168. [\[CrossRef\]](#)
11. Chen, W.; Shang, G.; Ji, A.; Zhou, C.; Wang, X.; Xu, C.; Li, Z.; Hu, K. An Overview on Visual SLAM: From Tradition to Semantic. *Remote Sens.* **2022**, *14*, 3010. [\[CrossRef\]](#)
12. Zhu, J.; Li, H.; Zhang, T. Camera, LiDAR, and IMU Based Multi-Sensor Fusion SLAM: A Survey. *Tsinghua Sci. Technol.* **2023**, *29*, 415–429. [\[CrossRef\]](#)
13. DeTone, D.; Malisiewicz, T.; Rabinovich, A. SuperPoint: Self-Supervised Interest Point Detection and Description. In Proceedings of the 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Salt Lake City, UT, USA, 18–22 June 2018; pp. 337–33712. [\[CrossRef\]](#)
14. Lindenberger, P.; Sarlin, P.-E.; Pollefeys, M. LightGlue: Local Feature Matching at Light Speed. In Proceedings of the 2023 IEEE/CVF International Conference on Computer Vision (ICCV 2023), Paris, France, 2–6 October 2023; pp. 17581–17592. [\[CrossRef\]](#)

15. Godard, C.; Mac Aodha, O.; Firman, M.; Brostow, G. Digging Into Self-Supervised Monocular Depth Estimation. In Proceedings of the 2019 IEEE/CVF International Conference on Computer Vision (ICCV 2019), Seoul, South Korea, 27 October–2 November 2019; pp. 3827–3837. [\[CrossRef\]](#)
16. Teed, Z.; Deng, J. DROID-SLAM: Deep Visual SLAM for Monocular, Stereo, and RGB-D Cameras. *arXiv* **2021**. [\[CrossRef\]](#)
17. Luo, H.; Liu, Y.; Guo, C.; Li, Z.; Song, W. SuperVINS: A Real-Time Visual-Inertial SLAM Framework for Challenging Imaging Conditions. *IEEE Sens. J.* **2025**, *25*, 26042–26050. [\[CrossRef\]](#)
18. Chen, L.-C.; Zhu, Y.; Papandreou, G.; Schroff, F.; Adam, H. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. *arXiv* **2018**. [\[CrossRef\]](#)
19. Li, S.; Zhang, D.; Xian, Y.; Li, B.; Zhang, T.; Zhong, C. Overview of Deep Learning Application on Visual SLAM. *Displays* **2022**, *74*, 102298. [\[CrossRef\]](#)
20. Martínez-Otzeta, J.M.; Rodríguez-Moreno, I.; Mendiáldua, I.; Sierra, B. RANSAC for Robotic Applications: A Survey. *Sensors* **2023**, *23*, 327. [\[CrossRef\]](#) [\[PubMed\]](#)
21. Chen, W.; Zhou, C.; Shang, G.; Wang, X.; Li, Z.; Xu, C.; Hu, K. SLAM Overview: From Single Sensor to Heterogeneous Fusion. *Remote Sens.* **2022**, *14*, 6033. [\[CrossRef\]](#)
22. Gonçalves, A.; Pereira, T.; Lopes, D.; Cunha, F.; Lopes, F.; Coutinho, F.; Barreiros, J.; Durães, J.; Santos, P.; Simões, F.; et al. Enhancing Nut-Tightening Processes in the Automotive Industry: Integration of 3D Vision Systems with Collaborative Robots. *Automation* **2025**, *6*, 8. [\[CrossRef\]](#)
23. Pereira, T.; Santos, V.; Gameiro, T.; Viegas, C.; Ferreira, N. Evaluation of Different Filtering Methods Devoted to Magnetometer Data Denoising. *Electronics* **2024**, *13*, 2006. [\[CrossRef\]](#)
24. Peng, Z.; Cheng, S.; Li, X.; Li, K.; Cai, M.; You, L. Dynamic Visual SLAM Integrated with IMU for Unmanned Scenarios. In Proceedings of the 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC 2022), Macau, China, 8–12 October 2022; pp. 4247–4253. [\[CrossRef\]](#)
25. Jia, G.; Li, X.; Zhang, D.; Xu, W.; Lv, H.; Shi, Y.; Cai, M. Visual-SLAM Classical Framework and Key Techniques: A Review. *Sensors* **2022**, *22*, 4582. [\[CrossRef\]](#)
26. Huang, Y.; Huang, B.; Zhang, Z.; Shi, Y.; Yuan, Y.; Sun, J. Loop Closure Detection Method Based on Similarity Differences between Image Blocks. *Sensors* **2023**, *23*, 8632. [\[CrossRef\]](#)
27. Álvarez-Tuñón, O.; Brodskiy, Y.; Kayacan, E. Monocular Visual Simultaneous Localization and Mapping: (R)Evolution From Geometry to Deep Learning-Based Pipelines. *IEEE Trans. Artif. Intell.* **2024**, *5*, 1990–2010. [\[CrossRef\]](#)
28. Loo, S.Y.; Amiri, A.J.; Mashohor, S.; Tang, S.H.; Zhang, H. CNN-SVO: Improving the Mapping in Semi-Direct Visual Odometry Using Single-Image Depth Prediction. In Proceedings of the 2019 IEEE International Conference on Robotics and Automation (ICRA 2019), Montreal, QC, Canada, 20–24 May 2019; pp. 5218–5223. [\[CrossRef\]](#)
29. Sánchez, J.; Monzón, N.; Salgado, A. An Analysis and Implementation of the Harris Corner Detector. *Image Process. Line* **2018**, *8*, 305–328. [\[CrossRef\]](#)
30. Shi, J.; Tomasi, C. Good Features to Track. In Proceedings of the 1994 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 1994), Seattle, WA, USA, 21–23 June 1994; pp. 593–600. [\[CrossRef\]](#)
31. Trajković, M.; Hedley, M. Fast Corner Detection. *Image Vis. Comput.* **1998**, *16*, 75–87. [\[CrossRef\]](#)
32. Lindeberg, T. Scale Invariant Feature Transform. *Scholarpedia* **2012**, *7*, 10491. [\[CrossRef\]](#)
33. Bay, H.; Tuytelaars, T.; Van Gool, L. SURF: Speeded up Robust Features. In *Computer Vision—ECCV 2006*; Lecture Notes in Computer Science; Springer: Berlin/Heidelberg, Germany, 2006; pp. 404–417.
34. Calonder, M.; Lepetit, V.; Strecha, C.; Fua, P. BRIEF: Binary Robust Independent Elementary Features. In Proceedings of the European Conference on Computer Vision (ECCV 2010), Heraklion, Crete, Greece, 5–11 September 2010; pp. 778–792.
35. Rublee, E.; Rabaud, V.; Konolige, K.; Bradski, G. ORB: An Efficient Alternative to SIFT or SURF. In Proceedings of the 2011 International Conference on Computer Vision (ICCV 2011), Barcelona, Spain, 6–13 November 2011; pp. 2564–2571. [\[CrossRef\]](#)
36. Jakubovic, A.; Velagic, J. Image Feature Matching and Object Detection Using Brute-Force Matchers. In Proceedings of the 2018 International Symposium ELMAR (ELMAR 2018), Zadar, Croatia, 16–18 September 2018; pp. 83–86. [\[CrossRef\]](#)
37. Muja, M.; Lowe, D. Fast Approximate Nearest Neighbors with Automatic Algorithm Configuration. In Proceedings of the 4th International Conference on Computer Vision Theory and Applications (VISAPP 2009), Lisbon, Portugal, 5–8 February 2009; Volume 1, pp. 331–340.
38. Li, S.; Xu, C.; Xie, M. A Robust O(n) Solution to the Perspective-n-Point Problem. *IEEE Trans. Pattern Anal. Mach. Intell.* **2012**, *34*, 1444–1450. [\[CrossRef\]](#) [\[PubMed\]](#)
39. Zhang, J.; Yao, Y.; Deng, B. Fast and Robust Iterative Closest Point. *IEEE Trans. Pattern Anal. Mach. Intell.* **2021**, *44*, 1. [\[CrossRef\]](#) [\[PubMed\]](#)
40. Zhan, Z.; Xu, H.; Fang, Z.; Wei, X.; Hu, Y.; Wang, C. Bundle Adjustment in the Eager Mode. *arXiv* **2024**. [\[CrossRef\]](#)

41. Kümmerle, R.; Grisetti, G.; Strasdat, H.; Konolige, K.; Burgard, W. g2o: A General Framework for Graph Optimization. In Proceedings of the 2011 IEEE International Conference on Robotics and Automation (ICRA 2011), Shanghai, China, 9–13 May 2011; pp. 3607–3613. [\[CrossRef\]](#)
42. Dellaert, F. Factor Graphs and GTSAM: A Hands-on Introduction. Technical Report, Georgia Institute of Technology, Atlanta, GA, USA, 2012. Available online: <https://repository.gatech.edu/server/api/core/bitstreams/b3606eb4-ce55-4c16-8495-767bd46f0351/content>.
43. Klein, G.; Murray, D. Parallel Tracking and Mapping for Small AR Workspaces. In Proceedings of the 6th IEEE/ACM International Symposium on Mixed and Augmented Reality (ISMAR 2007), Nara, Japan, 13–16 November 2007; pp. 225–234. [\[CrossRef\]](#)
44. Mur-Artal, R.; Montiel, J.M.M.; Tardós, J.D. ORB-SLAM: A Versatile and Accurate Monocular SLAM System. *IEEE Trans. Robot.* **2015**, *31*, 1147–1163. [\[CrossRef\]](#)
45. Mur-Artal, R.; Tardós, J.D. ORB-SLAM2: An Open-Source SLAM System for Monocular, Stereo, and RGB-D Cameras. *IEEE Trans. Robot.* **2017**, *33*, 1255–1262. [\[CrossRef\]](#)
46. Campos, C.; Elvira, R.; Gómez Rodríguez, J.J.; Montiel, J.M.M.; Tardós, J.D. ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM. *IEEE Trans. Robot.* **2021**, *37*, 1874–1890. [\[CrossRef\]](#)
47. Perdices, E.; López, L.M.; Cañas, J.M. LineSLAM: Visual Real-Time Localization Using Lines and UKF. In *Advances in Intelligent Systems and Computing*; Springer: Cham, Switzerland, 2013; pp. 663–678. [\[CrossRef\]](#)
48. Chen, Z.; Sheng, W.; Yang, G.; Su, Z.; Liang, B. Comparison and Analysis of Feature Method and Direct Method in Visual SLAM Technology for Social Robots. In Proceedings of the WCICA 2018, Changsha, China, 4–8 July 2018; pp. 413–417. [\[CrossRef\]](#)
49. He, X.; Gao, W.; Sheng, C.; Zhang, Z.; Pan, S.; Duan, L.; Zhang, H.; Lu, X. LiDAR-Visual-Inertial Odometry Based on Optimized Visual Point-Line Features. *Remote Sens.* **2022**, *14*, 622. [\[CrossRef\]](#)
50. Newcombe, R.A.; Lovegrove, S.; Davison, A.J. DTAM: Dense Tracking and Mapping in Real-Time. In Proceedings of the 2011 International Conference on Computer Vision (ICCV 2011), Barcelona, Spain, 6–13 November 2011; pp. 2320–2327.
51. Engel, J.; Schöps, T.; Cremers, D. LSD-SLAM: Large-Scale Direct Monocular SLAM. In *Computer Vision—ECCV 2014*; Lecture Notes in Computer Science; Springer: Cham, Switzerland, 2014; pp. 834–849. [\[CrossRef\]](#)
52. Engel, J.; Sturm, J.; Cremers, D. Semi-Dense Visual Odometry for a Monocular Camera. In Proceedings of the 2013 IEEE International Conference on Computer Vision (ICCV 2013), Sydney, Australia, 1–8 December 2013; pp. 1449–1456. [\[CrossRef\]](#)
53. Lebedev, E.; Uchaev, D.; Uchaev, D. Stability assessment and performance comparison of VSLAM frameworks on open indoor datasets. *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.* **2024**, *XLVIII-2/W5-2024*, 101–107. [\[CrossRef\]](#)
54. Engel, J.; Koltun, V.; Cremers, D. Direct sparse odometry. *IEEE Trans. Pattern Anal. Mach. Intell.* **2017**, *40*, 611–625. [\[CrossRef\]](#) [\[PubMed\]](#)
55. Cruz, L.; Lucio, D.; Velho, L. Kinect and RGBD Images: Challenges and Applications. In Proceedings of the SIBGRAPI-T, Ouro Preto, Brazil, 22–25 August 2012. [\[CrossRef\]](#)
56. Newcombe, R.A.; Izadi, S.; Hilliges, O.; Molyneaux, D.; Kim, D.; Davison, A.J.; Kohi, P.; Shotton, J.; Hodges, S.; Fitzgibbon, A. KinectFusion: Real-Time Dense Surface Mapping and Tracking. In Proceedings of the 10th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2011), Basel, Switzerland, 26–29 October 2011. [\[CrossRef\]](#)
57. Whelan, T.; Salas-Moreno, R.F.; Glocker, B.; Davison, A.J.; Leutenegger, S. ElasticFusion: Real-time dense SLAM and light source estimation. *Int. J. Robot. Res.* **2016**, *35*, 1697–1716. [\[CrossRef\]](#)
58. Whelan, T.; Kaess, M.; Fallon, M.F.; Johannsson, H.; Leonard, J.J.; McDonald, J.B. Kintinuous: Spatially Extended KinectFusion. In Proceedings of the Twenty-Sixth AAAI Conference on Artificial Intelligence (AAAI-12), Toronto, ON, Canada, 22–26 July 2012.
59. Li, Y.; Yunus, R.; Brasch, N.; Navab, N.; Tombari, F. RGB-D SLAM with Structural Regularities. In Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA 2020), Paris, France, 31 May–4 June 2020; pp. 11581–11587.
60. Park, S.; Schöps, T.; Pollefeys, M. Illumination Change Robustness in Direct Visual SLAM. In Proceedings of the 2017 IEEE International Conference on Robotics and Automation (ICRA 2017), Singapore, 29 May–3 June 2017; pp. 4523–4530.
61. Huang, J.; Zhao, S.; Zhang, L. E2-VINS: An Event-Enhanced Visual-Inertial SLAM Scheme for Dynamic Environments. *Appl. Sci.* **2025**, *15*, 1314. [\[CrossRef\]](#)
62. Cioffi, G.; Scaramuzza, D. Tightly-Coupled Fusion of Global Positional Measurements in Optimization-Based Visual-Inertial Odometry. In Proceedings of the 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2020), Las Vegas, NV, USA, 25–29 October 2020; pp. 5089–5095. [\[CrossRef\]](#)
63. Munguía, R.; Nuño, E.; Aldana, C.I.; Urzua, S. A Visual-Aided Inertial Navigation and Mapping System. *Int. J. Adv. Robot. Syst.* **2016**, *13*, 1. [\[CrossRef\]](#)
64. Palezieux, N.; Nägeli, T.; Hilliges, O. Duo-VIO: Fast, Light-Weight, Stereo Inertial Odometry. In Proceedings of the 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2016), Daejeon, South Korea, 9–14 October 2016; pp. 2237–2242.
65. Wang, A.; Zhong, R.; Zheng, K.; Fang, H. LL-SLAM: Lightweight Loosely-Coupled Visual-Inertial SLAM. In Proceedings of the 2023 42nd Chinese Control Conference (CCC 2023), Tianjin, China, 23–25 July 2023; pp. 3632–3637.

66. Lu, J.; Shen, Y.; Hang, Q.; Tang, Y. Research on Tightly Coupled Visual Inertial SLAM Based on Point and Line Features. *Multimed. Syst.* **2024**, *30*, 286. [\[CrossRef\]](#)
67. Mourikis, A.I.; Roulletiotis, S.I. A Multi-State Constraint Kalman Filter for Vision-Aided Inertial Navigation. In Proceedings of the 2007 IEEE International Conference on Robotics and Automation (ICRA 2007), Rome, Italy, 10–14 April 2007; pp. 3565–3572.
68. Leutenegger, S.; Lynen, S.; Bosse, M.; Siegwart, R.; Furgale, P. Keyframe-based visual-inertial odometry using nonlinear optimization. *Int. J. Rob. Res.* **2014**, *34*, 314–334. [\[CrossRef\]](#)
69. Leutenegger, S. OKVIS2: Realtime Scalable Visual-Inertial SLAM with Loop Closure. *arXiv* **2022**, arXiv:2202.09199. [\[CrossRef\]](#)
70. Qin, T.; Li, P.; Shen, S. VINS-Mono: A Robust and Versatile Monocular Visual-Inertial State Estimator. *IEEE Trans. Robot.* **2017**, *34*, 1004–1020. [\[CrossRef\]](#)
71. Liu, C.; Wang, T.; Li, Z.; Tian, P. A Novel Real-Time Autonomous Localization Algorithm Based on Weighted Loosely-Coupled Visual-Inertial Data of the Velocity Layer. *Appl. Sci.* **2025**, *15*, 989.
72. Geiger, A.; Lenz, P.; Stiller, C.; Urtasun, R. Vision Meets Robotics: The KITTI Dataset. *Int. J. Robot. Res.* **2013**, *32*, 1231–1237. [\[CrossRef\]](#)
73. Burri, M.; Nikolic, J.; Gohl, P.; Schneider, T.; Rehder, J.; Omari, S.; Achtelik, M.; Siegwart, R.Y. The EuRoC Micro Aerial Vehicle Datasets. *Int. J. Robot. Res.* **2016**, *35*, 1157–1163.
74. Schubert, D.; Goll, T.; Demmel, N.; Usenko, V.C.; Stückler, J.; Cremers, D. The TUM VI Benchmark for Evaluating Visual-Inertial Odometry. In Proceedings of the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2018), Madrid, Spain, 1–5 October 2018; pp. 1680–1687.
75. Pire, T.; Mujica, M.; Civera, J.; Kofman, E. The Rosario dataset: Multisensor data for localization and mapping in agricultural environments. *Int. J. Robot. Res.* **2018**, *38*, 633–641. [\[CrossRef\]](#)
76. Polvara, R.; Molina, S.; Hroob, I.; Papadimitriou, A.; Tsiolis, K.; Giakoumis, D.; Likothanassis, S.D.; Tzovaras, D.; Cielniak, G.; Hanheide, M. Bacchus Long-Term (BLT) data set: Acquisition of the agricultural multimodal BLT data set with automated robot deployment. *J. Field Robot.* **2023**, *41*, 2280–2298. [\[CrossRef\]](#)
77. Marzoa Tanco, M.; Trinidad Barnech, G.; Andrade, F.; Baliosian, J.; Llofriu, M.; Di Martino, J.M.; Tejera, G. MAgro dataset: A dataset for simultaneous localization and mapping in agricultural environments. *Int. J. Robot. Res.* **2023**, *43*, 591–601. [\[CrossRef\]](#)
78. Jiang, P.; Osteen, P.R.; Wigness, M.B.; Saripalli, S. RELIS-3D Dataset: Data, Benchmarks and Analysis. In Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA 2020), Paris, France, 31 May–4 June 2020; pp. 1110–1116.
79. Ali, I.; Durmush, A.; Suominen, O.J.; Yli-Hietanen, J.; Peltonen, S.; Collin, J.; Gotchev, A.P. FinnForest dataset: A forest landscape for visual SLAM. *Robot. Auton. Syst.* **2020**, *132*, 103610. [\[CrossRef\]](#)
80. Ort, T.; Gilitschenski, I.; Rus, D. GROUNDED: A localizing ground penetrating radar evaluation dataset for learning to localize in inclement weather. *Int. J. Robot. Res.* **2023**, *42*, 901–916. [\[CrossRef\]](#)
81. Schmidt, F.; Blessing, C.; Enzweiler, M.; Valada, A. ROVER: A Multi-Season Dataset for Visual SLAM. *arXiv* **2024**, arXiv:2412.02506. [\[CrossRef\]](#)
82. Jia, Y.; Wang, S.; Shao, S.; Wang, Y.; Zhang, F.-Y.; Wang, T. M2UD: A Multi-model, Multi-scenario, Uneven-terrain Dataset for Ground Robot with Localization and Mapping Evaluation. *arXiv* **2025**, arXiv:2503.12387. [\[CrossRef\]](#)
83. Chen, C.; Wang, B.; Lu, C.X.; Trigoni, N.; Markham, A. Deep Learning for Visual Localization and Mapping: A Survey. *arXiv* **2023**, arXiv:2308.14039. [\[CrossRef\]](#)
84. Mokssit, S.; Bonilla Licea, D.; Guermah, B.; Ghogho, M. Deep Learning Techniques for Visual SLAM: A Survey. *IEEE Access* **2023**, *11*, 20026–20050. [\[CrossRef\]](#)
85. Liao, Y.; Di, Y.; Zhu, K.; Zhou, H.; Lu, M.; Zhang, Y.; Duan, Q.; Liu, J. Local feature matching from detector-based to detector-free: A survey. *Appl. Intell.* **2024**, *54*, 3954–3989. [\[CrossRef\]](#)
86. Liu, C.; Xu, J.; Wang, F. A Review of Keypoints' Detection and Feature Description in Image Registration. *Sci. Program.* **2021**, *2021*, 8509164:1–8509164:25. [\[CrossRef\]](#)
87. Dusmanu, M.; Rocco, I.; Pajdla, T.; Pollefeys, M.; Sivic, J.; Torii, A.; Sattler, T. D2-Net: A Trainable CNN for Joint Description and Detection of Local Features. In Proceedings of the 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2019), Long Beach, CA, USA, 15–20 June 2019; pp. 8084–8093.
88. Tian, Y.; Balntas, V.; Ng, T.; Barroso Laguna, A.; Demiris, Y.; Mikolajczyk, K. D2D: Keypoint Extraction with Describe to Detect Approach. In Proceedings of the Asian Conference on Computer Vision (ACCV 2020), Kyoto, Japan, 30 November–4 December 2020.
89. Li, Z.; Ma, J. Learning Feature Matching via Matchable Keypoint-Assisted Graph Neural Network. *IEEE Trans. Image Process.* **2023**, *34*, 154–169. [\[CrossRef\]](#)
90. Yi, K.M.; Trulls, E.; Lepetit, V.; Fua, P. LIFT: Learned Invariant Feature Transform. In Proceedings of the European Conference on Computer Vision (ECCV 2016), Amsterdam, The Netherlands, 11–14 October 2016.

91. Sarlin, P.-E.; DeTone, D.; Malisiewicz, T.; Rabinovich, A. SuperGlue: Learning Feature Matching with Graph Neural Networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2020), Seattle, WA, USA, 14–19 June 2020; pp. 4937–4946.
92. Zhao, X.; Wu, X.; Miao, J.; Chen, W.; Chen, P.C.Y.; Li, Z. ALIKE: Accurate and Lightweight Keypoint Detection and Descriptor Extraction. *IEEE Trans. Multimed.* **2021**, *25*, 3101–3112. [\[CrossRef\]](#)
93. Xie, T.; Dai, K.; Wang, K.; Li, R.; Zhao, L. DeepMatcher: A Deep Transformer-Based Network for Robust and Accurate Local Feature Matching. *arXiv* **2023**, arXiv:2301.02993. [\[CrossRef\]](#)
94. Rocco, I.; Cimpoi, M.; Arandjelović, R.; Torii, A.; Pajdla, T.; Sivic, J. NCNet: Neighbourhood Consensus Networks for Estimating Image Correspondences. *IEEE Trans. Pattern Anal. Mach. Intell.* **2020**, *44*, 1020–1034. [\[CrossRef\]](#) [\[PubMed\]](#)
95. Li, X.; Han, K.; Li, S.; Prisacariu, V.A. Dual-Resolution Correspondence Networks. *arXiv* **2020**, arXiv:2006.08844. [\[CrossRef\]](#)
96. Teed, Z.; Deng, J. RAFT: Recurrent All-Pairs Field Transforms for Optical Flow. *arXiv* **2020**, arXiv:2003.12039. [\[CrossRef\]](#)
97. Efe, U.; Ince, K.G.; Alatan, A.A. DFM: A Performance Baseline for Deep Feature Matching. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Nashville, TN, USA, 19–25 June 2021; pp. 4279–4288.
98. Kwiatkowski, M.; Matern, S.; Hellwich, O. ConDL: Detector-Free Dense Image Matching. *arXiv* **2024**, arXiv:2408.02766.
99. Zhang, H.; Shafiq, M.O. Survey of Transformers and Towards Ensemble Learning Using Transformers for Natural Language Processing. *J. Big Data* **2024**, *11*, 25. [\[CrossRef\]](#) [\[PubMed\]](#)
100. Jiang, W.; Trulls, E.; Hosang, J.H.; Tagliasacchi, A.; Yi, K.M. COTR: Correspondence Transformer for Matching Across Images. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2021), Montreal, QC, Canada, 11–17 October 2021; pp. 6187–6197.
101. Sun, J.; Shen, Z.; Wang, Y.; Bao, H.; Zhou, X. LoFTR: Detector-Free Local Feature Matching with Transformers. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2021), Nashville, TN, USA, 20–25 June 2021; pp. 8918–8927.
102. Bökman, G.; Kahl, F. A Case for Using Rotation-Invariant Features in State-of-the-Art Feature Matchers. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2022), New Orleans, LA, USA, 19–20 June 2022; pp. 5106–5115.
103. Wang, Q.; Zhang, J.; Yang, K.; Peng, K.; Stiefelhagen, R. MatchFormer: Interleaving Attention in Transformers for Feature Matching. In Proceedings of the Asian Conference on Computer Vision (ACCV 2022), Macau, China, 4–8 December 2022.
104. He, X.; Yu, H.; Peng, S.; Tan, D.; Shen, Z.; Bao, H.; Zhou, X. MatchAnything: Universal Cross-Modality Image Matching with Large-Scale Pre-Training. *arXiv* **2025**, arXiv:2501.07556.
105. Khan, F.; Salahuddin, S.; Javidnia, H. Deep Learning-Based Monocular Depth Estimation Methods—A State-of-the-Art Review. *Sensors* **2020**, *20*, 2272. [\[CrossRef\]](#) [\[PubMed\]](#)
106. Eigen, D.; Puhrsch, C.; Fergus, R. Depth Map Prediction from a Single Image Using a Multi-Scale Deep Network. In Proceedings of the Neural Information Processing Systems (NIPS 2014), Montreal, QC, Canada, 8–13 December 2014.
107. Laina, I.; Rupprecht, C.; Belagiannis, V.; Tombari, F.; Navab, N. Deeper Depth Prediction with Fully Convolutional Residual Networks. In Proceedings of the 2016 Fourth International Conference on 3D Vision (3DV 2016), Stanford, CA, USA, 25–28 October 2016; pp. 239–248.
108. Teed, Z.; Deng, J. DeepV2D: Video to Depth with Differentiable Structure from Motion. *arXiv* **2018**, arXiv:1812.04605.
109. Zhou, M.; Yu, H.; Li, Z.; Zhang, Y. Self-Supervised Monocular Depth Estimation Based on Differential Attention. *Algorithms* **2025**, *18*, 590. [\[CrossRef\]](#)
110. Zhou, T.; Brown, M.A.; Snavely, N.; Lowe, D.G. Unsupervised Learning of Depth and Ego-Motion from Video. In Proceedings of the 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2017), Honolulu, HI, USA, 21–26 July 2017; pp. 6612–6619. Available online: <https://api.semanticscholar.org/CorpusID:11977588> (accessed on 5 October 2025).
111. Lyu, X.; Liu, L.; Wang, M.; Kong, X.; Liu, L.; Liu, Y.; Chen, X.; Yuan, Y. HR-Depth: High Resolution Self-Supervised Monocular Depth Estimation. In Proceedings of the Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI-20), New York, NY, USA, 7–12 February 2020. Available online: <https://api.semanticscholar.org/CorpusID:229152988> (accessed on 5 October 2025).
112. Yan, J.; Zhao, H.; Bu, P.; Jin, Y. Channel-Wise Attention-Based Network for Self-Supervised Monocular Depth Estimation. In Proceedings of the 2021 International Conference on 3D Vision (3DV 2021), London, UK, 1–3 December 2021; pp. 464–473. Available online: <https://api.semanticscholar.org/CorpusID:245501910> (accessed on 10 November 2025).
113. Shim, D.; Kim, H.J. SwinDepth: Unsupervised Depth Estimation using Monocular Sequences via Swin Transformer and Densely Cascaded Network. In Proceedings of the 2023 IEEE International Conference on Robotics and Automation (ICRA 2023), London, UK, 29 May–2 June 2023; pp. 4983–4990. Available online: <https://api.semanticscholar.org/CorpusID:255942422> (accessed on 5 October 2025).

114. Tang, S.; Lu, T.; Liu, X.; Zhou, H.; Zhang, Y. CATNet: Convolutional attention and transformer for monocular depth estimation. *Pattern Recognit.* **2023**, *145*, 109982. Available online: <https://api.semanticscholar.org/CorpusID:262097131> (accessed on 5 October 2025). [CrossRef]
115. Liu, P.; Zhang, Z.; Meng, Z.; Gao, N. Transformer-based monocular depth estimation with hybrid attention fusion and progressive regression. *Neurocomputing* **2024**, *620*, 129268. Available online: <https://api.semanticscholar.org/CorpusID:275051548> (accessed on 18 November 2025). [CrossRef]
116. Li, H.; Zhang, Z.; Hao, Z.; Xu, B.; Wang, W.; Sun, J. PAR-mono: Monocular video depth estimation network based on channel separation and dynamic attention. *Multimed. Syst.* **2024**, *31*, 6. Available online: <https://api.semanticscholar.org/CorpusID:274674968> (accessed on 18 November 2025). [CrossRef]
117. Wang, S.; Clark, R.; Wen, H.; Trigoni, A. DeepVO: Towards End-to-End Visual Odometry with Deep Recurrent Convolutional Neural Networks. In Proceedings of the 2017 IEEE International Conference on Robotics and Automation (ICRA 2017), Singapore, 29 May–3 June 2017; pp. 2043–2050. Available online: <https://api.semanticscholar.org/CorpusID:9114952> (accessed on 18 November 2025).
118. Clark, R.; Wang, S.; Wen, H.; Markham, A.; Trigoni, A. VINet: Visual-Inertial Odometry as a Sequence-to-Sequence Learning Problem. *arXiv* **2017**, arXiv:1701.08376. Available online: <https://api.semanticscholar.org/CorpusID:17112447> (accessed on 22 November 2025).
119. Li, R.; Wang, S.; Long, Z.; Gu, D. UnDeepVO: Monocular Visual Odometry Through Unsupervised Deep Learning. In Proceedings of the 2018 IEEE International Conference on Robotics and Automation (ICRA 2018), Brisbane, QLD, Australia, 21–25 May 2018; pp. 7286–7291. Available online: <https://api.semanticscholar.org/CorpusID:206853077> (accessed on 22 November 2025).
120. Almalioglu, Y.; Saputra, M.R.U.; Buarque de Gusmão, P.P.; Markham, A.; Trigoni, A. GANVO: Unsupervised Deep Monocular Visual Odometry and Depth Estimation with Generative Adversarial Networks. In Proceedings of the 2019 IEEE International Conference on Robotics and Automation (ICRA 2019), Montreal, QC, Canada, 20–24 May 2019; pp. 5474–5480. Available online: <https://api.semanticscholar.org/CorpusID:52282758> (accessed on 23 November 2025).
121. Yang, N.; von Stumberg, L.; Wang, R.; Cremers, D. D3VO: Deep Depth, Deep Pose and Deep Uncertainty for Monocular Visual Odometry. In Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2020), Seattle, WA, USA, 14–19 June 2020; pp. 1278–1289. Available online: <https://api.semanticscholar.org/CorpusID:211677466> (accessed on 23 November 2025).
122. Han, L.; Lin, Y.; Du, G.; Lian, S. DeepVIO: Self-Supervised Deep Learning of Monocular Visual-Inertial Odometry Using 3D Geometric Constraints. In Proceedings of the 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2019), Macau, China, 3–8 November 2019; pp. 6906–6913. Available online: <https://api.semanticscholar.org/CorpusID:195699451> (accessed on 24 November 2025).
123. Shamwell, J.; Leung, S.; Nothwang, W.D. Vision-Aided Absolute Trajectory Estimation Using an Unsupervised Deep Network with Online Error Correction. In Proceedings of the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2018), Madrid, Spain, 1–5 October 2018; pp. 2524–2531. Available online: <https://api.semanticscholar.org/CorpusID:3895695> (accessed on 26 November 2025).
124. Li, R.; Wang, S.; Gu, D. DeepSLAM: A Robust Monocular SLAM System with Unsupervised Deep Learning. *IEEE Trans. Ind. Electron.* **2021**, *68*, 3577–3587. Available online: <https://api.semanticscholar.org/CorpusID:216294549> (accessed on 26 November 2025). [CrossRef]
125. Bruno, H.M.S.; Colombini, E. LIFT-SLAM: A deep-learning feature-based monocular visual SLAM method. *Neurocomputing* **2020**, *455*, 97–110. Available online: <https://api.semanticscholar.org/CorpusID:232478547> (accessed on 27 November 2025). [CrossRef]
126. Peng, X.; Liu, Z.; Li, W.; Tan, P.; Cho, S.I.; Wang, Q. DVI-SLAM: A Dual Visual Inertial SLAM Network. In Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA), 2023; pp. 12020–12026. Available online: <https://api.semanticscholar.org/CorpusID:262459196> (accessed on 27 November 2025).
127. Sun, Y.; Wang, Q.; Yan, C.; Feng, Y.; Tan, R.; Shi, X.; Wang, X. D-VINS: Dynamic Adaptive Visual-Inertial SLAM with IMU Prior and Semantic Constraints in Dynamic Scenes. *Remote Sens.* **2023**, *15*, 3881. [CrossRef]
128. Zhao, W.; Sun, H.; Ma, S.; Wang, H. LET-SE2-VINS: A Hybrid Optical Flow Framework for Robust Visual-Inertial SLAM. *Sensors* **2025**, *25*, 3837. [CrossRef]
129. Wang, S.; Hu, Q.; Zhang, X.; Li, W.; Wang, Y.; Zheng, E. LVID-SLAM: A Lightweight Visual-Inertial SLAM for Dynamic Scenes Based on Semantic Information. *Sensors* **2025**, *25*, 4117. [CrossRef]
130. Chen, W.; Li, Y.; Tian, Z.; Zhang, F. 2D and 3D object detection algorithms from images: A Survey. *Array* **2023**, *19*, 100305. [CrossRef]
131. Pereira, T.; Gameiro, T.; Pedro, J.; Viegas, C.; Ferreira, N.M.F. Vision System for a Forestry Navigation Machine. *Sensors* **2024**, *24*, 1475. [CrossRef] [PubMed]
132. Girshick, R.B. Fast R-CNN. In Proceedings of the IEEE International Conference on Computer Vision (ICCV 2015), Santiago, Chile, 7–13 December 2015. Available online: <https://api.semanticscholar.org/CorpusID:206770307> (accessed on 27 November 2025).

133. Liu, W.; Anguelov, D.; Erhan, D.; Szegedy, C.; Reed, S.E.; Fu, C.-Y.; Berg, A.C. SSD: Single Shot MultiBox Detector. In Proceedings of the European Conference on Computer Vision (ECCV 2016), Amsterdam, The Netherlands, 11–14 October 2016. Available online: <https://api.semanticscholar.org/CorpusID:2141740> (accessed on 29 November 2025).
134. Redmon, J.; Divvala, S.K.; Girshick, R.B.; Farhadi, A. You Only Look Once: Unified, Real-Time Object Detection. In Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2016), Las Vegas, NV, USA, 27–30 June 2016; pp. 779–788. [CrossRef]
135. Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *arXiv* **2020**, arXiv:2010.11929. [CrossRef]
136. Touvron, H.; Cord, M.; Douze, M.; Massa, F.; Sablayrolles, A.; Jégou, H. Training Data-Efficient Image Transformers and Distillation through Attention. In Proceedings of the International Conference on Machine Learning (ICML 2020), Vienna, Austria, 13–18 July 2020. Available online: <https://api.semanticscholar.org/CorpusID:229363322> (accessed on 29 November 2025).
137. Wang, W.; Xie, E.; Li, X.; Fan, D.-P.; Song, K.; Liang, D.; Lu, T.; Luo, P.; Shao, L. Pyramid Vision Transformer: A Versatile Backbone for Dense Prediction without Convolutions. In Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV 2021), Montreal, QC, Canada, 11–17 October 2021; pp. 548–558. Available online: <https://api.semanticscholar.org/CorpusID:232035922> (accessed on 29 November 2025).
138. Liu, Z.; Lin, Y.; Cao, Y.; Hu, H.; Wei, Y.; Zhang, Z.; Lin, S.; Guo, B. Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows. In Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV 2021), Montreal, QC, Canada, 11–17 October 2021; pp. 9992–10002. Available online: <https://api.semanticscholar.org/CorpusID:232352874> (accessed on 29 November 2025).
139. Xie, E.; Wang, W.; Yu, Z.; Anandkumar, A.; Álvarez, J.M.; Luo, P. SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers. In Proceedings of the Neural Information Processing Systems (NeurIPS 2021), Virtual Conference (Sydney, Australia), 6–14 December 2021. Available online: <https://api.semanticscholar.org/CorpusID:235254713> (accessed on 29 November 2025).
140. Kirillov, A.; Mintun, E.; Ravi, N.; Mao, H.; Rolland, C.; Gustafson, L.; Xiao, T.; Whitehead, S.; Berg, A.C.; Lo, W.-Y.; et al. Segment Anything. In Proceedings of the 2023 IEEE/CVF International Conference on Computer Vision (ICCV 2023), Paris, France, 2–6 October 2023; pp. 3992–4003. Available online: <https://api.semanticscholar.org/CorpusID:257952310> (accessed on 29 November 2025).
141. Cheng, B.; Collins, M.D.; Zhu, Y.; Liu, T.; Huang, T.S.; Adam, H.; Chen, L.-C. Panoptic-DeepLab: A Simple, Strong, and Fast Baseline for Bottom-Up Panoptic Segmentation. In Proceedings of the 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2020), Seattle, WA, USA, 14–19 June 2020; pp. 12472–12482. Available online: <https://api.semanticscholar.org/CorpusID:208248153> (accessed on 30 November 2025).
142. Cheng, B.; Misra, I.; Schwing, A.G.; Kirillov, A.; Girdhar, R. Masked-attention Mask Transformer for Universal Image Segmentation. In Proceedings of the 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022), New Orleans, LA, USA, 19–24 June 2022; pp. 1280–1289. Available online: <https://api.semanticscholar.org/CorpusID:244799297> (accessed on 30 November 2025).
143. Zhi, S.; Laidlow, T.; Leutenegger, S.; Davison, A.J. In-Place Scene Labelling and Understanding with Implicit Scene Representation. In Proceedings of the 2021 IEEE/CVF International Conference on Computer Vision (ICCV 2021), Virtual Conference (Montreal, QC, Canada), 11–17 October 2021; pp. 15818–15827. Available online: <https://api.semanticscholar.org/CorpusID:232417682> (accessed on 30 November 2025).
144. Chou, Z.-T.; Huang, S.-Y.; Liu, I.-J.; Wang, Y.-C.F. GSNeRF: Generalizable Semantic Neural Radiance Fields with Enhanced 3D Scene Understanding. In Proceedings of the 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024), Seattle, WA, USA, 17–21 June 2024; pp. 20806–20815. Available online: <https://api.semanticscholar.org/CorpusID:268253324> (accessed on 30 November 2025).
145. Keetha, N.V.; Karhade, J.; Jatavallabhula, K.M.; Yang, G.; Scherer, S.; Ramanan, D.; Luiten, J. SplatTAM: Splat, Track & Map 3D Gaussians for Dense RGB-D SLAM. In Proceedings of the 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024), Seattle, WA, USA, 17–21 June 2024; pp. 21357–21366. Available online: <https://api.semanticscholar.org/CorpusID:265609060> (accessed on 30 November 2025).
146. Lisus, D.; Holmes, C.T. Towards Open World NeRF-Based SLAM. In Proceedings of the 2023 20th Conference on Robots and Vision (CRV 2023), Montreal, QC, Canada, 6–8 June 2023; pp. 37–44. Available online: <https://api.semanticscholar.org/CorpusID:255545865> (accessed on 30 November 2025).
147. Gameiro, T.; Pereira, T.; Viegas, C.; Di Giorgio, F.; Ferreira, N.F. Robots for Forest Maintenance. *Forests* **2024**, *15*, 381. [CrossRef]
148. Pereira, T.; Gameiro, T.; Viegas, C.; Santos, V.; Ferreira, N. Sensor Integration in a Forestry Machine. *Sensors* **2023**, *23*, 9853. [CrossRef] [PubMed]

149. Yu, Y.; Zhang, K.; Liu, H.; Yang, L.; Zhang, D. Real-Time Visual Localization of the Picking Points for a Ridge-Planting Strawberry Harvesting Robot. *IEEE Access* **2020**, *8*, 116556–116568. [\[CrossRef\]](#)
150. Giubilato, R.; Stürzl, W.; Wedler, A.; Triebel, R. Challenges of SLAM in Extremely Unstructured Environments: The DLR Planetary Stereo, Solid-State LiDAR, Inertial Dataset. *IEEE Robot. Autom. Lett.* **2022**, *7*, 10078–10085. [\[CrossRef\]](#)
151. Castanheiro, L.F.; Tommaselli, A.M.G.; Hyyti, H.; Kukko, A. Experimental assessment of IMU data processing techniques for a backpack mobile laser scanning system. *Int. Arch. Photogramm. Remote Sens. Spat. Inf. Sci.* **2024**, *XLVIII-3-2024*, 57–62. [\[CrossRef\]](#)
152. Cristóvão, M.P.; Portugal, D.; Carvalho, A.E.; Ferreira, J.F. A LiDAR-Camera-Inertial-GNSS Apparatus for 3D Multimodal Dataset Collection in Woodland Scenarios. *Sensors* **2023**, *23*, 6676. [\[CrossRef\]](#) [\[PubMed\]](#)
153. Islam, R.; Habibullah, H.; Hossain, T. AGRI-SLAM: A real-time stereo visual SLAM for agricultural environment. *Auton. Robot.* **2023**, *47*, 649–668. [\[CrossRef\]](#)
154. Zhang, X.; Yao, G.; Vishwakarma, S.; Dalin, C.; Komarek, A.M.; Kanter, D.R.; Davis, K.F.; Pfeifer, K.J.; Zhao, J.; Zou, T.; et al. Quantitative assessment of agricultural sustainability reveals divergent priorities among nations. *One Earth* **2021**, *4*, 1262–1277. [\[CrossRef\]](#)
155. Sanyaolu, M.; Sadowski, A. The Role of Precision Agriculture Technologies in Enhancing Sustainable Agriculture. *Sustainability* **2024**, *16*, 6668. [\[CrossRef\]](#)
156. Sharma, A.; Jain, A.; Gupta, P.; Chowdary, V. Machine Learning Applications for Precision Agriculture: A Comprehensive Review. *IEEE Access* **2021**, *9*, 4843–4873. [\[CrossRef\]](#)
157. Tsiakas, K.; Papadimitriou, A.; Pechlivani, E.M.; Giakoumis, D.; Frangakis, N.; Gasteratos, A.; Tzovaras, D. An autonomous navigation framework for holonomic mobile robots in confined agricultural environments. *Robotics* **2023**, *12*, 146. [\[CrossRef\]](#)
158. Chen, M.; Tang, Y.; Zou, X.; Huang, Z.; Zhou, H.; Chen, S. 3D global mapping of large-scale unstructured orchard integrating eye-in-hand stereo vision and SLAM. *Comput. Electron. Agric.* **2021**, *187*, 106237. [\[CrossRef\]](#)
159. Aghi, D.; Mazzia, V.; Chiaberge, M. Local Motion Planner for Autonomous Navigation in Vineyards with a RGB-D Camera-Based Algorithm and Deep Learning Synergy. *arXiv* **2020**, arXiv:2005.12815. [\[CrossRef\]](#)
160. Cremona, J.; Civera, J.; Kofman, E.; Pire, T. GNSS-stereo-inertial SLAM for arable farming. *J. Field Robot.* **2023**, *41*, 2215–2225. [\[CrossRef\]](#)
161. Song, K.; Qiu, R.; Yang, G.; Li, J. Monocular Visual-Inertial Odometry for Agricultural Environments. *IEEE Access* **2022**, *10*, 103975–103986. [\[CrossRef\]](#)
162. Tran, Q.D.; Hoang, Q.H.; Kim, G.-W. Robust Monocular Visual-Inertial Odometry for Agricultural Vehicles Based on IMU-Augmented 3D Feature Point Correction. *IEEE Access* **2025**, *13*, 132611–132623. [\[CrossRef\]](#)
163. Feng, H.; Zhao, G.; Liu, C.; Wang, Y.; Wang, J. Motion-Coupled Mapping Algorithm for Hybrid Rice Canopy. *arXiv* **2025**, arXiv:2502.16134. [\[CrossRef\]](#)
164. Gameiro, T.; Pereira, T.; Moghadaspoura, H.; Di Giorgio, F.; Viegas, C.; Ferreira, N.; Ferreira, J.; Soares, S.; Valente, A. Evaluation of PID-Based Algorithms for UGVs. *Algorithms* **2025**, *18*, 63. [\[CrossRef\]](#)
165. Heupel, C.; Wolf, P.; Berns, K. Tree-SLAM: Localization and Mapping in Dense Forest Environments for Autonomous Vehicles. In *Commercial Vehicle Technology 2024*; Springer: Wiesbaden, Germany, 2023. [\[CrossRef\]](#)
166. Pierzchala, M.; Giguère, P.; Astrup, R. Mapping forests using an unmanned ground vehicle with 3D LiDAR and graph-SLAM. *Comput. Electron. Agric.* **2018**, *145*, 217–225. [\[CrossRef\]](#)
167. Fäitli, T.; Hyyppä, E.; Hyyti, H.; Hakala, T.; Kaartinen, H.; Kukko, A.; Muhojoki, J.; Hyyppä, J. Integration of a Mobile Laser Scanning System with a Forest Harvester for Accurate Localization and Tree Stem Measurements. *Remote Sens.* **2024**, *16*, 3292. [\[CrossRef\]](#)
168. Liu, H.; Xu, G.; Liu, B.; Li, Y.; Yang, S.; Tang, J.; Pan, K.; Xing, Y. A real time LiDAR-Visual-Inertial object level semantic SLAM for forest environments. *ISPRS J. Photogramm. Remote Sens.* **2025**, *219*, 71–90. [\[CrossRef\]](#)
169. Shi, K.; Lin, W. Dynamic visual SLAM algorithm for urban forest environments based on semantic segmentation. *Meas. Sci. Technol.* **2025**, *36*, 056312. [\[CrossRef\]](#)

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